

The Unseen AI Disruptions for Power Grids: Large Language Model Induced Transients

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Edmonton, Alberta
200+ undergraduate programs
500+ graduate programs
40,000+ students from 160+ countries



From: <https://geology.com/canada/alberta.shtml>

Faculty of Engineering

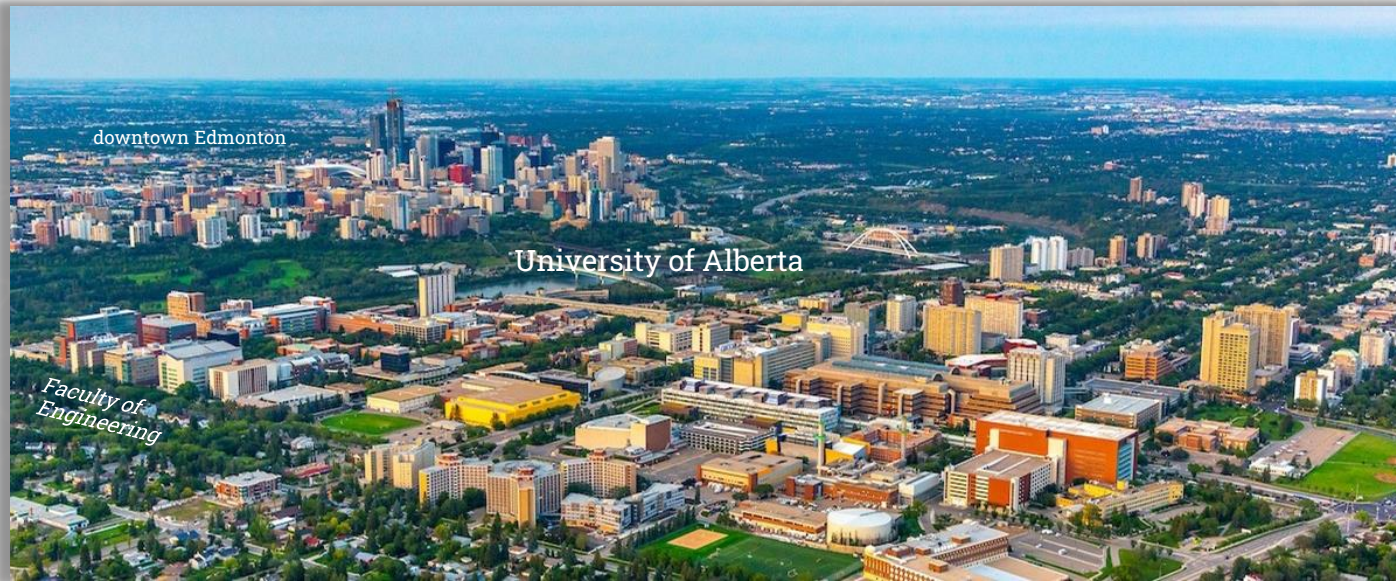
Top Engineering Schools in Canada
4400 undergraduate
1600 graduate students
220+ Professors



Donadeo Innovation Centre of Engineering (ICE)



ECE research facility (ECERF)



Overview of the north campus and downtown Edmonton

Outline

Background

Characteristics of AI Loads

Power Consumption Modeling of AI Workloads

AI Power Consumption Analysis: Case Studies

Future Research Directions

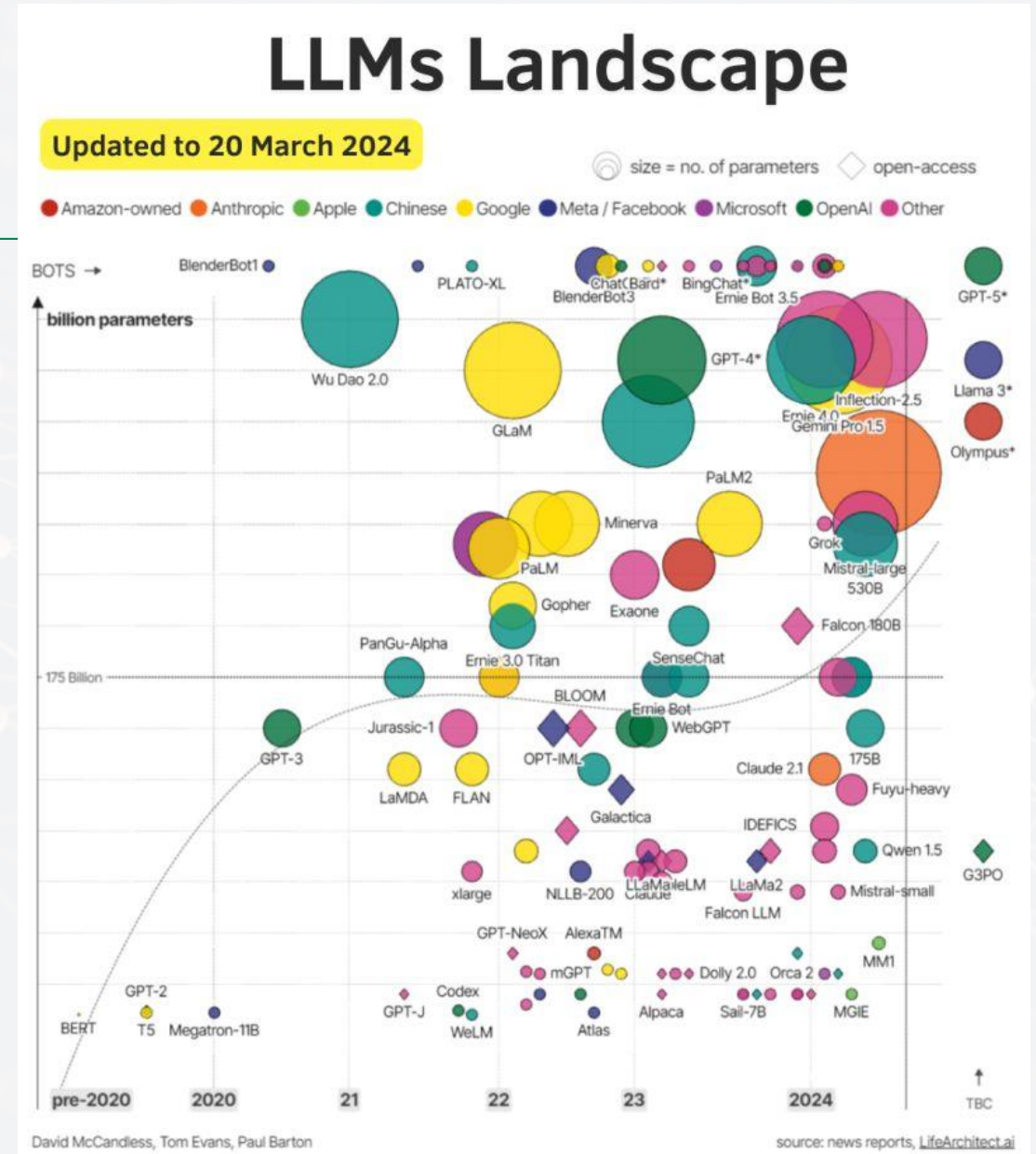
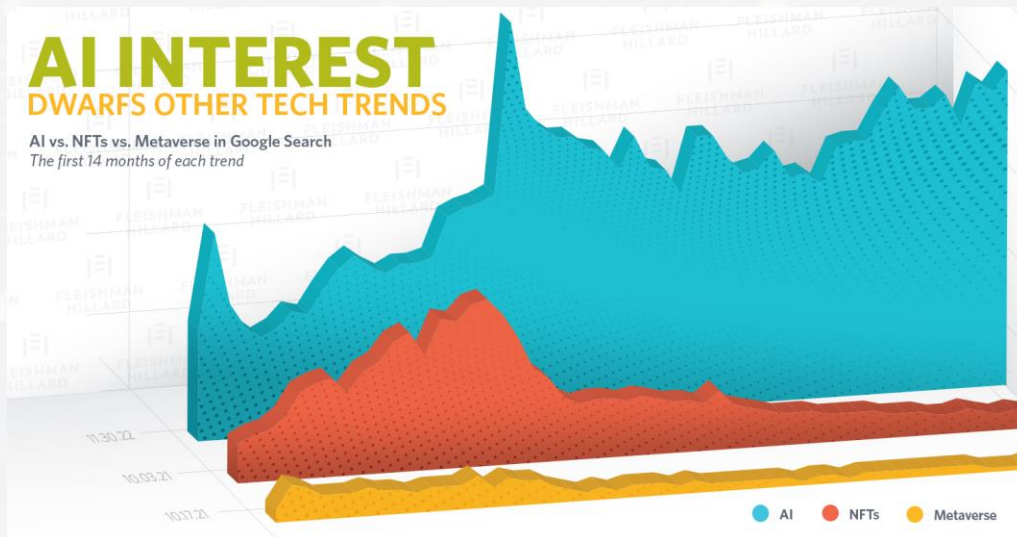
Conclusions



Booming AI

The 4th time wave of AI

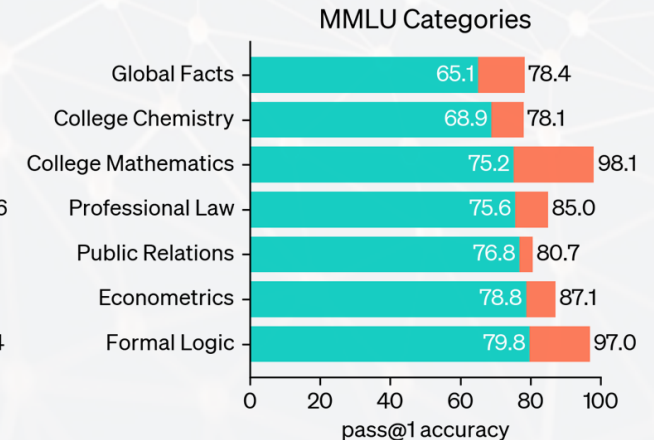
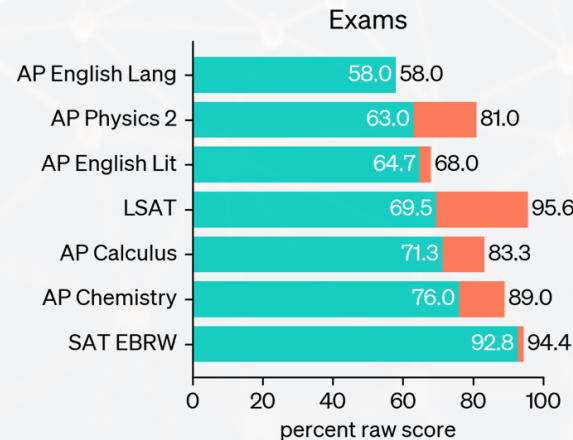
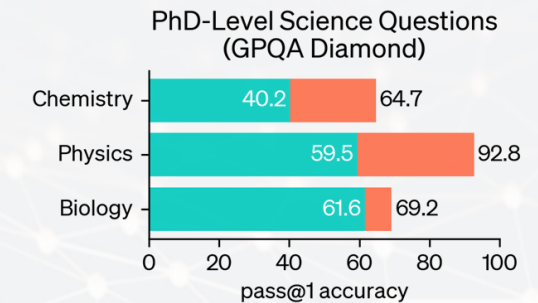
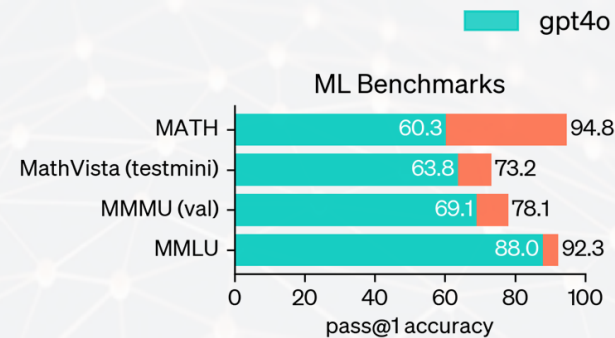
- AI tech. draw a great public attention.
- The tech cycle is in its beginning stage.
- AI impacts most of the society, as general purpose technology.



GenAI Evolving fast

New model can reason

- OpenAI o1 uses a chain of thought for problem-solving, similar to how a human thinks before responding to a question.
- Through reinforcement learning, o1 improves its reasoning, refines strategies, and learns from mistakes.
- o1 breaks down complex steps into simpler ones and adopts new approaches when needed.
- This process significantly enhances the model's reasoning ability.



GenAI Evolving fast

New model can reason – Solve OPF



I have three generators connected to bus 1,2,and 3 respectively, and these three buses are connected to each other, and transmission line limit is 2MW. The cost of these generators are \$1, \$3,\$5 per MW. Node 2 have a load of 5MW. Please solve the optimal power flow, and find the optimal dispatch.



Thought for 80 seconds ▾

To solve this optimal power flow problem, we'll analyze the given system consisting of three buses interconnected by transmission lines with a capacity limit of 2 MW each. The cost of generation at each bus is as follows:

- **Bus 1 (Generator 1):** \$1 per MW
- **Bus 2 (Generator 2):** \$3 per MW
- **Bus 3 (Generator 3):** \$5 per MW

Bus 2 has a load demand of 5 MW.

Objective: Minimize the total generation cost while meeting the load demand and not exceeding the transmission line limits.

Variables:

- G_1, G_2, G_3 : Generation at buses 1, 2, and 3, respectively (MW)
- P_{12}, P_{13}, P_{23} : Power flow between buses (MW)

Constraints:

1. Power Balance at Each Bus:

- **Bus 1:** $G_1 - P_{12} - P_{13} = 0$
- **Bus 2:** $G_2 + P_{12} - P_{23} = -5$ (since it's consuming 5 MW)
- **Bus 3:** $G_3 + P_{13} + P_{23} = 0$

2. Transmission Line Limits:

- $|P_{12}| \leq 2$
- $|P_{13}| \leq 2$
- $|P_{23}| \leq 2$

3. Generation Limits:

- $G_1 \geq 0$
- $G_2 \geq 0$
- $G_3 \geq 0$

4. Total Generation Equals Total Load:

- $G_1 + G_2 + G_3 = 5$

Objective Function:

Minimize $C = 1 \times G_1 + 3 \times G_2 + 5 \times G_3$

Solution Approach:

1. Attempt to Meet Load Without Violating Line Limits:

Since each line can carry a maximum of 2 MW, the maximum power that can be imported into Bus 2 is 4 MW (2 MW from Bus 1 and 2 MW from Bus 3). Therefore, Bus 2 must generate at least $5 - 4 = 1$ MW to meet its 5 MW demand.

Large Scale Deployment of Modern AI

Generative AI

- **Language** --- coding, translation, document processing, prediction, ...
- **Vision** --- pattern recognition, revising, detection, ...
- **Generalizable tasks**--- self driving, autonomous robotics, world model, ...



GenAI is Energy Hog

- Single GPT query consumes 1567% more energy than a Google Search query;
- Google's greenhouse gas emissions climbed nearly 50% over five years, mostly due to electricity powers AI data centers*;
- AI currently consumes 4.5 GW, with annual growth rate of 25% to 33%. By 2028, AI-related consumption could reach 14 GW to 18.7 GW**.

* Artificial intelligence's thirst for electricity, NPR July 10th 2024, <https://www.npr.org/2024/07/10/nx-s1-5028558/artificial-intelligences-thirst-for-electricity>

**The AI Disruption: Challenges and Guidance for Data Center Design, Schneider, https://www.se.com/ww/en/download/document/SPD_WP110_EN/

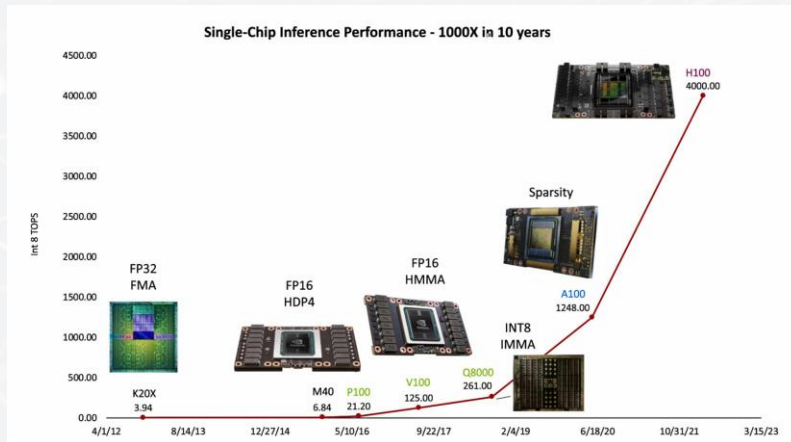
Type of Service	Estimated Energy Consumption (per query or operation)
Google Search Query	0.0003 kWh (1.08 kJ)
NLP/ChatGPT-4 Query	0.001-0.01 kWh (3.6-36 kJ) *
SQL Database Query	0.0001-0.001 kWh (0.36-3.6 kJ) **
Graph Database Query	0.0001-0.01 kWh (0.36-36 kJ) ***
Cloud Container	0.001-0.1 kWh (3.6-360 kJ) ****
Serverless Function	0.00001-0.001 kWh (0.036-3.6 kJ) *****

AI Development Bounded By Power and Energy

Low-latency LLMs inference on resource-constrained devices



Reliable LLMs services on **power/energy-constrained** infra.



Microsoft deal would reopen Three Mile Island nuclear plant to power AI

The owner of the shuttered Pennsylvania plant plans to bring it online by 2028, with the tech giant buying all the power it produces.

8 min 2200

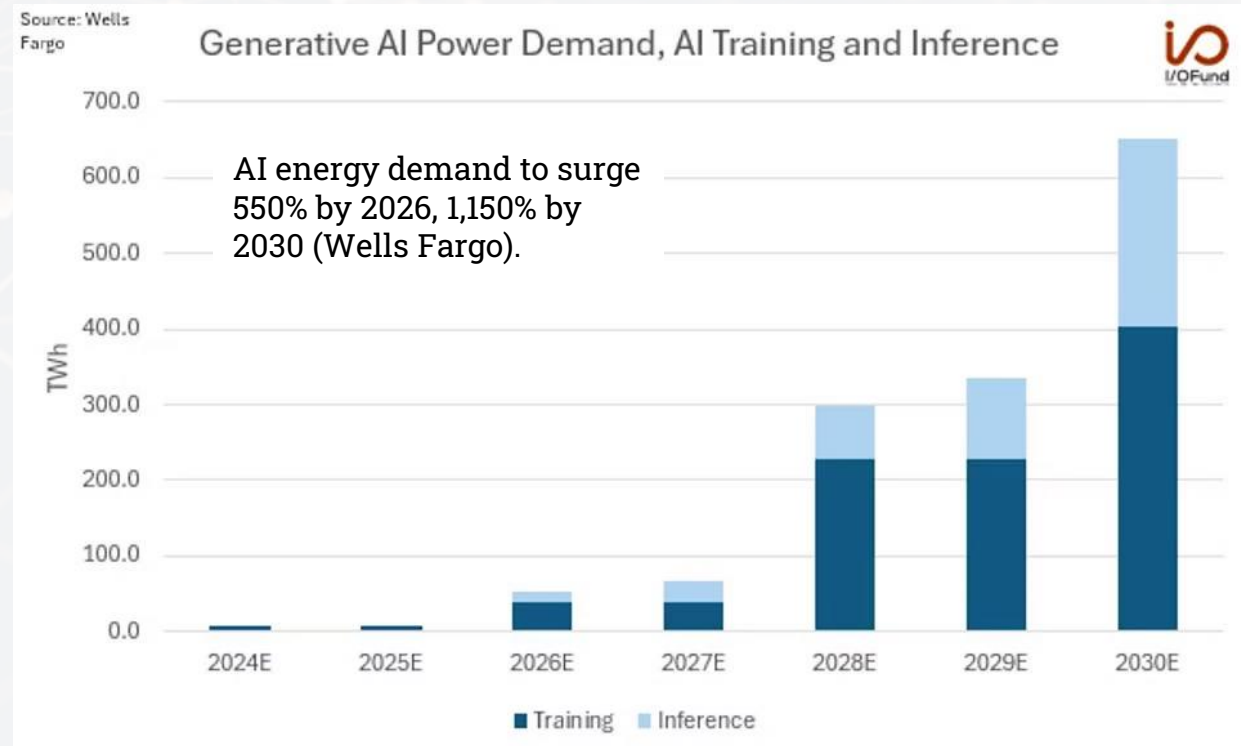


Three Mile Island was touted as the U.S.'s "worst nuclear reactor accident in history"

AI Electricity Demands

AI Electricity Demand Growth

- In 2030, 20-25% power consumption could be consumed by AI in the U.S., a significant increase from today's 4% or less (Wall Street Journal).
- Global data center power use to triple in 2024 (Morgan Stanley).
- Data center electricity consumption to more than double by 2030 (Electric Power Research Institute).
- Global electricity demand from AI, data centers, and crypto to reach 800 TWh by 2026 (IEA).



AI Electricity Demands

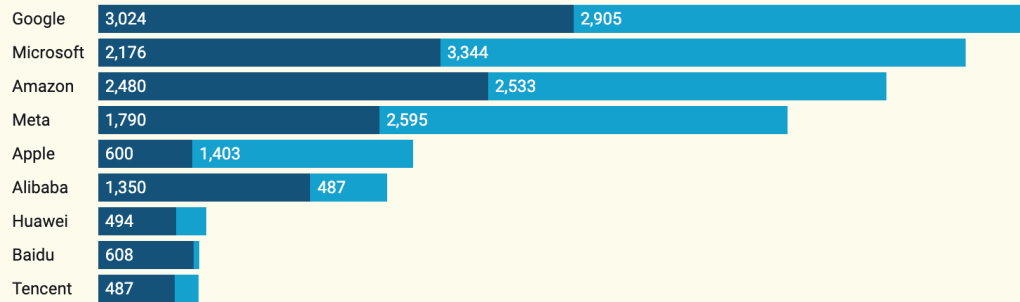
Google Cloud

Select a project ▼

← Llama 3.1

Estimated Hyperscaler Data Center Capacity (MW)

■ 2022 Capacity ■ Projected future capacity

Chart: Brian Potter • Source: [Semianalysis](#) • [Get the data](#) • Created with [Datawrapper](#)

Training Energy Use Training utilized a cumulative of 39.3M GPU hours of computation on H100-80GB (TDP of 700W) type hardware, per the table below. Training time is the total GPU time required for training each model and power consumption is the peak power capacity per GPU device used, adjusted for power usage efficiency.

Training Greenhouse Gas Emissions Estimated total location-based greenhouse gas emissions were 11,390 tons CO2eq for training. Since 2020, Meta has maintained net zero greenhouse gas emissions in its global operations and matched 100% of its electricity use with renewable energy, therefore the total market-based greenhouse gas emissions for training were 0 tons CO2eq.

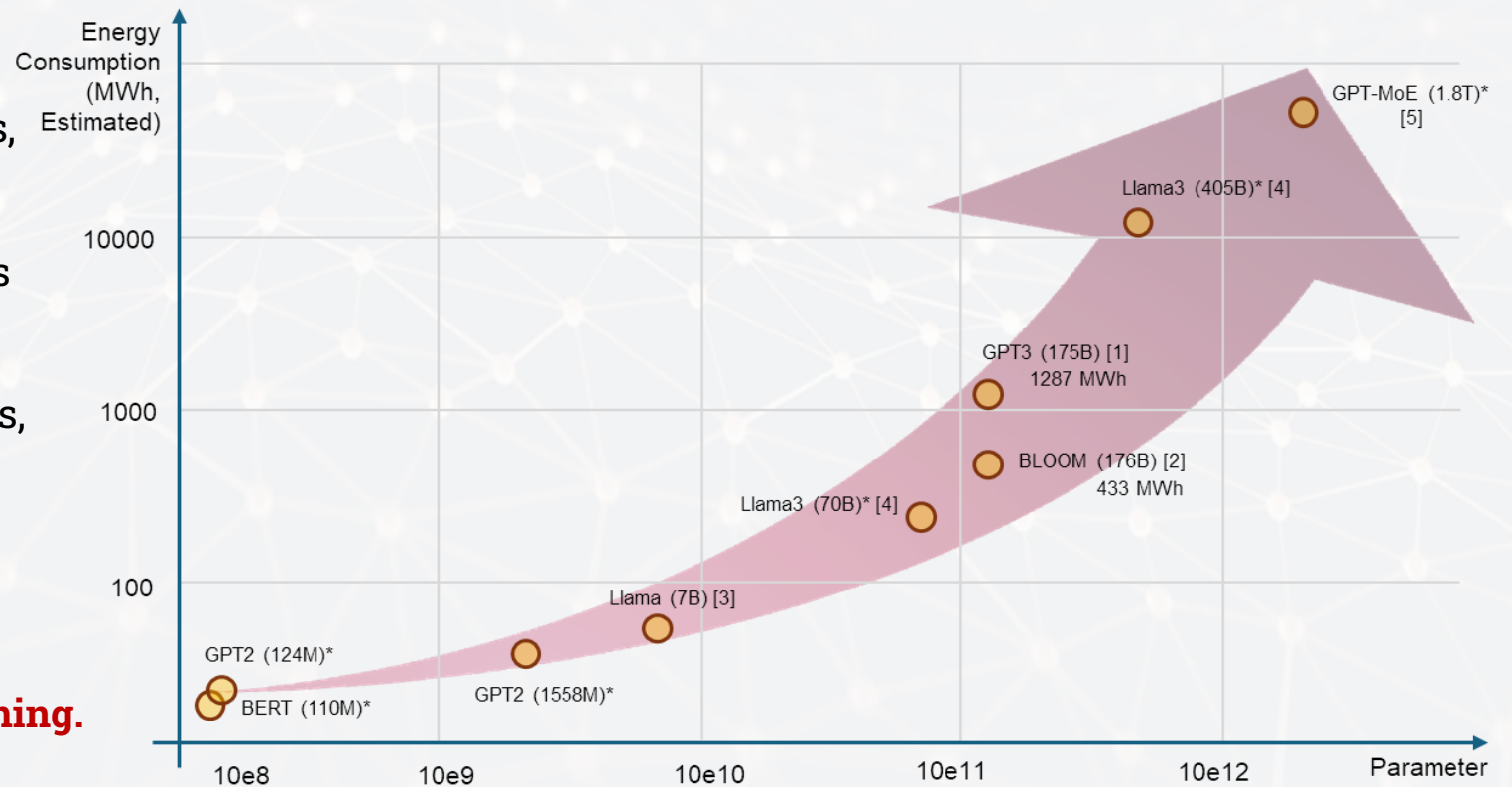
	Training Time (GPU hours)	Training Power Consumption (W)	Training Location-Based Greenhouse Gas Emissions(tons CO2eq)	Training Market-Based Greenhouse Gas Emissions(tons CO2eq)
Llama 3.1 8B	1.46M	700	420	0
Llama 3.1 70B	7.0M	700	2,040	0
Llama 3.1 405B	30.84M	700	8,930	0
Total	39.3M	-	11,390	0

AI is a Grid Challenger

Model Size Growth - Training Energy Consumption

- **LeNet-5** (1998) trained on 60k examples, ~60k parameters, single CPU
- **AlexNet** (2012) trained on 1.2M examples, ~60M parameters, dual GPUs
- **GPT3** (2020) trained on ~500B tokens, 175B parameters, 10e5 GPUs
- **GPT-MoE** (2023) trained on ~13T tokens, 1.8T parameters, 2.5*10e5 GPUs
- ... AI will become the largest single contributor of electricity demand increase in North America

We are already in GWh era in model training.
The challenge is not only the scale...

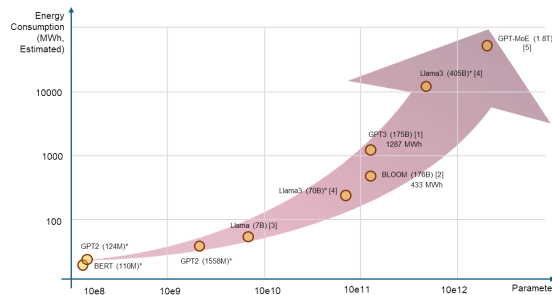


I - Background

AI is a Grid Challenger

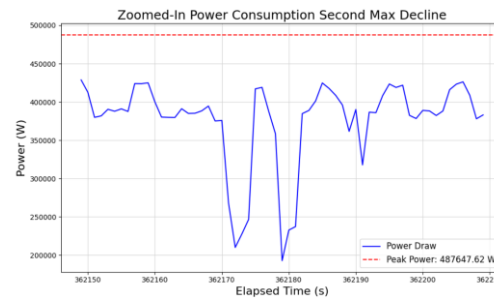
The Scale

- Train on power-intense data centers (now MW, future GW)
- 10,000-100,000 GPUs work as one (weeks to months)
- Huge **single point of load**, multiple locations in future



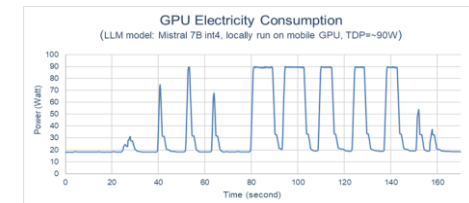
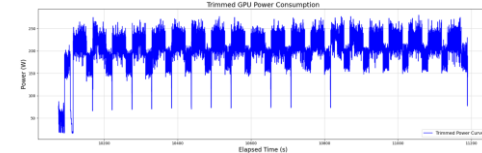
The Speed

- GPUs calculation (parallel, running in GHz, fast dynamics)
- High bandwidth data transfer (GB/s is slow, now is TB/s)
- Low latency is gold (faster in future)



The Chaos

- Unplanned stop (unpredictable)
- Sensitive to algorithm design (subjective)
- User behavior and question-related (stochastic)



Serving AI load Reliably

Traditional designs and operation of Power Grid may not fit well for AI Applications

Chip-level

- AI chip has higher power density
- Voltage Regulators and Power Management ICs
- Power Supply Units
- Step down voltages for CPUs, GPUs, memory
- High Current Delivery
- Low Ripple Voltage

Data-center-level

- AI server power density keeps increasing
- Power Distribution Units
- Uninterruptible Power Supplies
- Renewable Integration
- Energy Storage Systems
- Cooling Systems

Grid-level

- Increased Demand and Fluctuations (Grid Interface converter, Distribution Substation)
- Reliably, and sustainably
- Renewable Integration
- Voltage and Frequency
- Power Transience

The majority of companies working on AI **have not disclosed such data**, bringing additional challenges for accommodating and designing AI-oriented power infrastructure.

Outline

Background

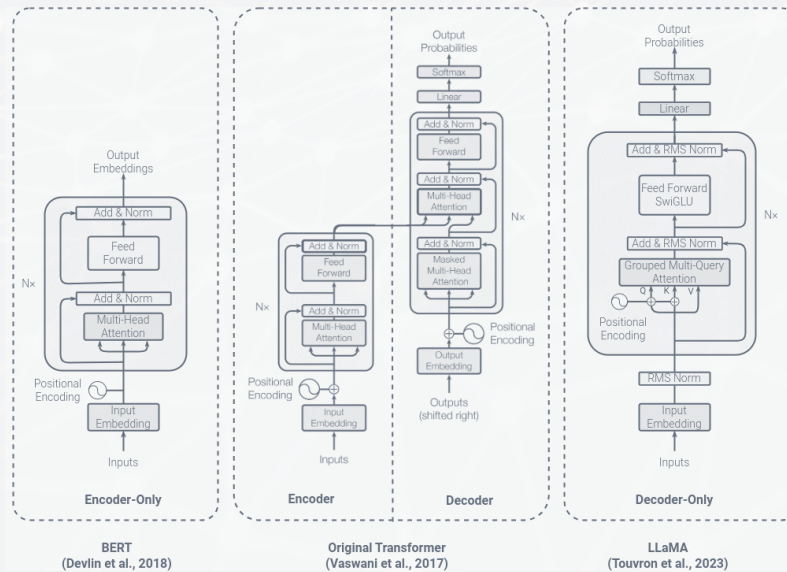
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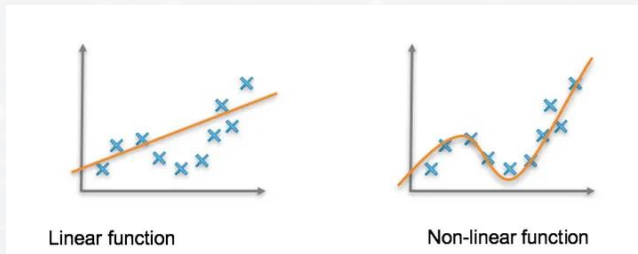
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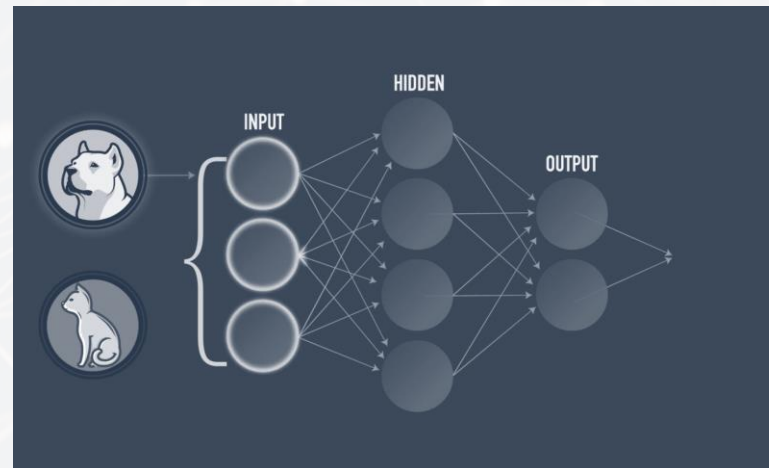
Conclusions



Neural Network But Becoming Huge



Statistical Model



Deep Neural Network



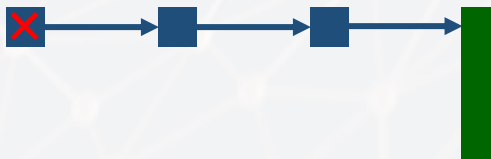
Large Language Model, text, voice, image, ...

The Hardware – Data Center system

There is no AI without Data Center now

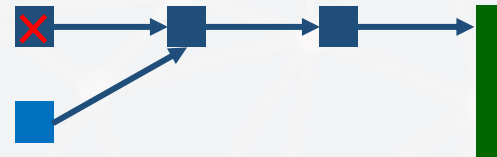
Tier I Basic Capacity

- A data center with a **single** path for power and cooling, and no backup components.
- Site-wide shutdowns are required for maintenance or repair work. Capacity or distribution failures will impact the site.
- Uptime of 99.671% (28.8 hours of downtime annually).



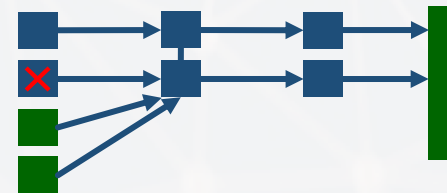
Tier II Redundant Capacity Components

- A data center with a **single** path for power and cooling, and some redundant and backup components.
- Site-wide shutdowns for maintenance are still required. Capacity failures may impact the site. Distribution failures will impact the site.
- Uptime of 99.741% (22 hours of downtime annually).



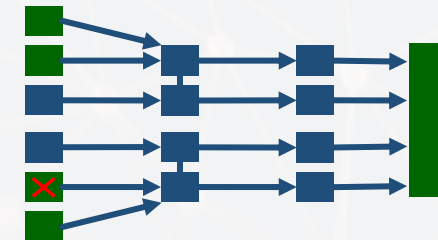
Tier III Concurrently Maintainable

- A data center with **multiple paths** for power and cooling, and redundant systems that allow the staff to work on the setup without taking it offline.
- The site is still exposed to an equipment failure or operator error.
- Uptime of 99.982% (1.6 hours of downtime annually).

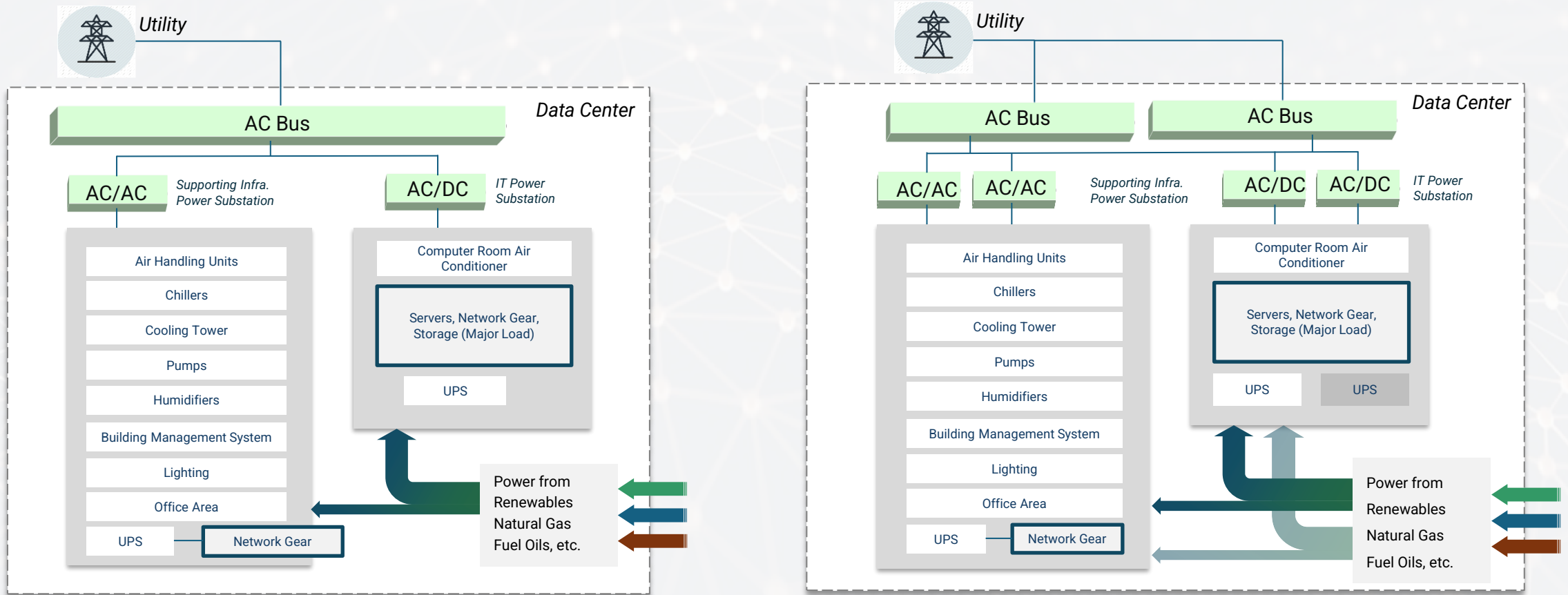


Tier IV Fault Tolerant

- A **completely** fault-tolerant data center with redundancy for every component.
- An individual equipment failure or distribution path interruption will not impact operations.
- A Fault Tolerant site is also Concurrently Maintainable.
- Uptime of 99.995% (26.3 minutes of downtime annually).



The Hardware – Data Center system



The Hardware – the server

DGX H100 max consumption: 10.2 kW;

- 8*H100 GPUs;
- About 9-10kW under full load;

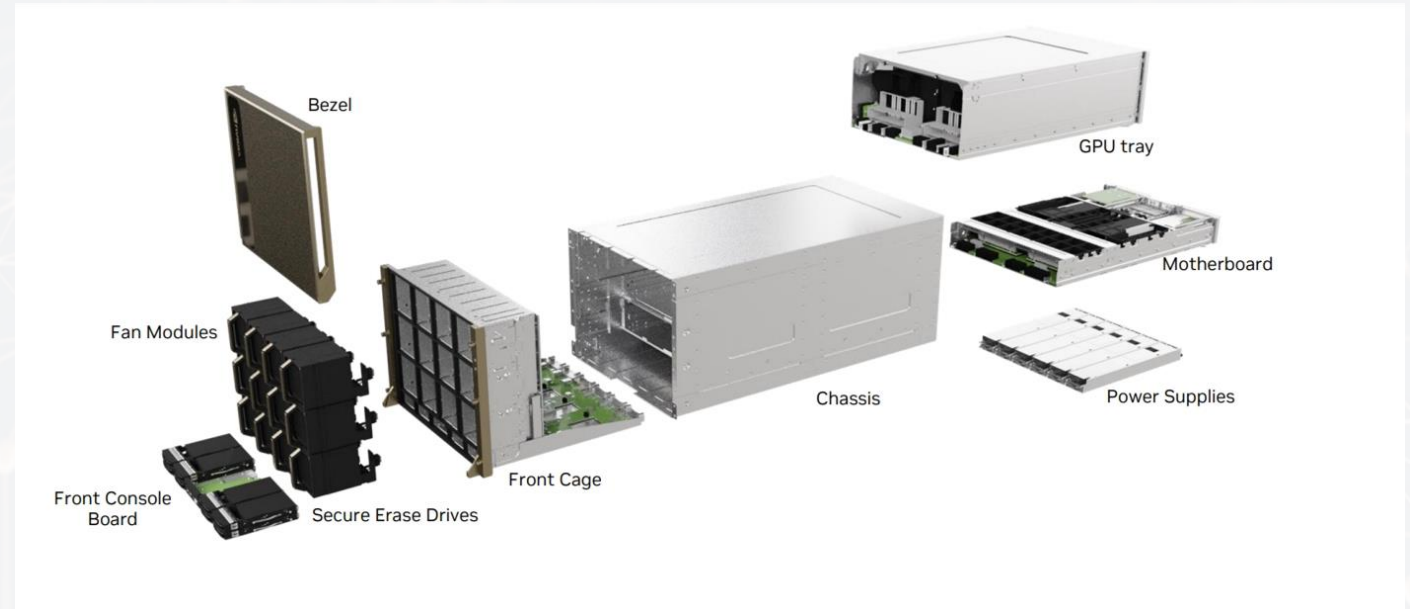
Model Training

Llama 3.1 70B took 7.0 million H100 hours*

- If targeted at 1 month training time, the single training task power is around **1MW**

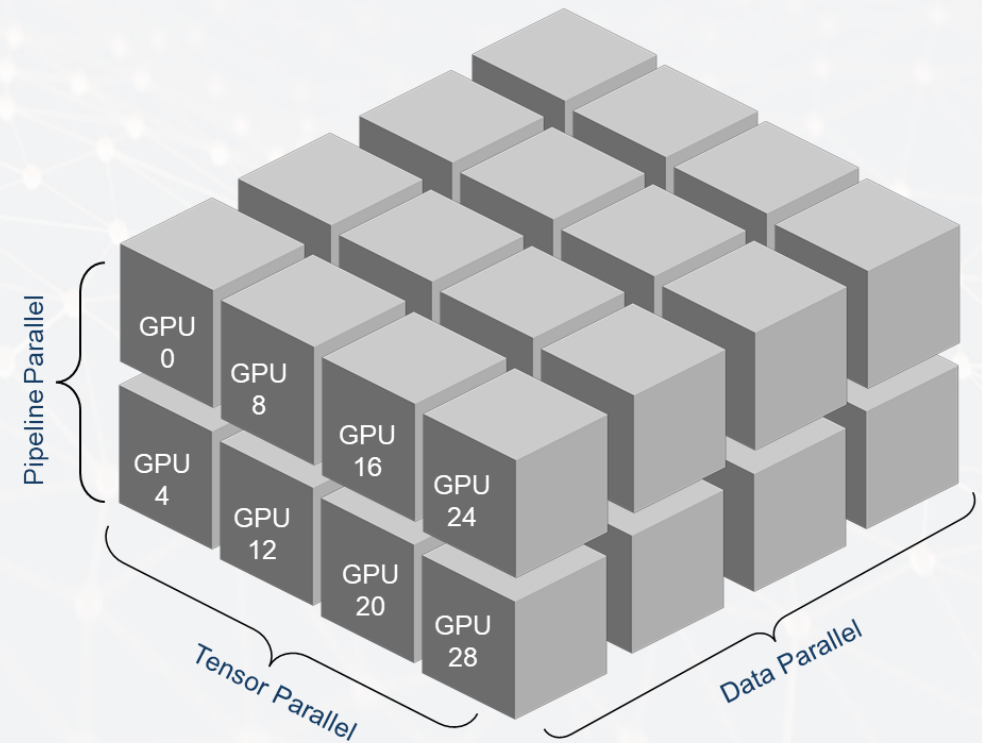
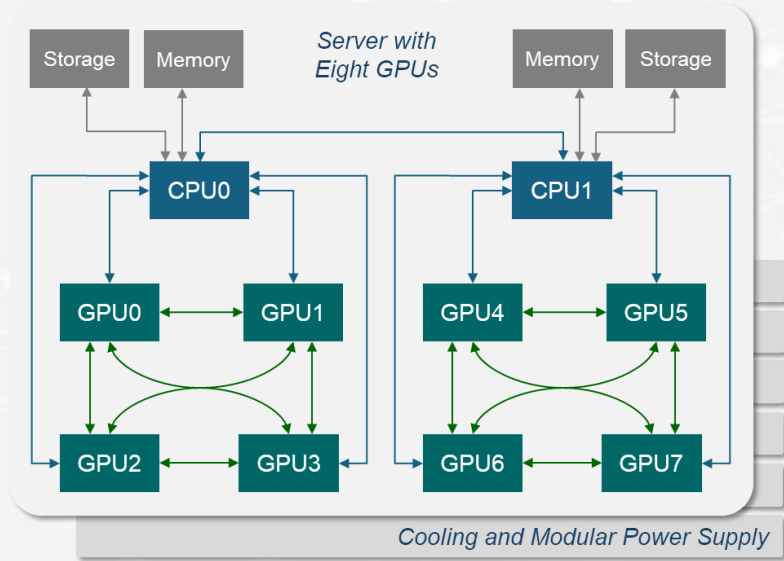
Model Inference

- Inside Meta, only 10% of GPUs are served for training tasks;
- Llama 3.1 405B needs over 10 A100s for inference, while serving one user request requires more than **12kW**.



DGX H100 Breakdown

The Hardware – the component



The Hardware – the design

AI-specific large, reliable power supply;

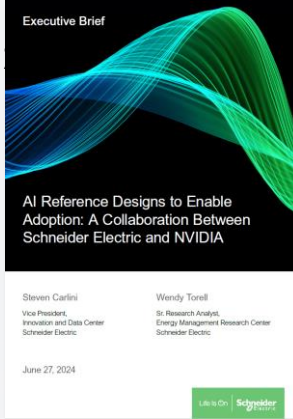
Moving towards Gigawatt data centers;

High reliability against service interruptions.

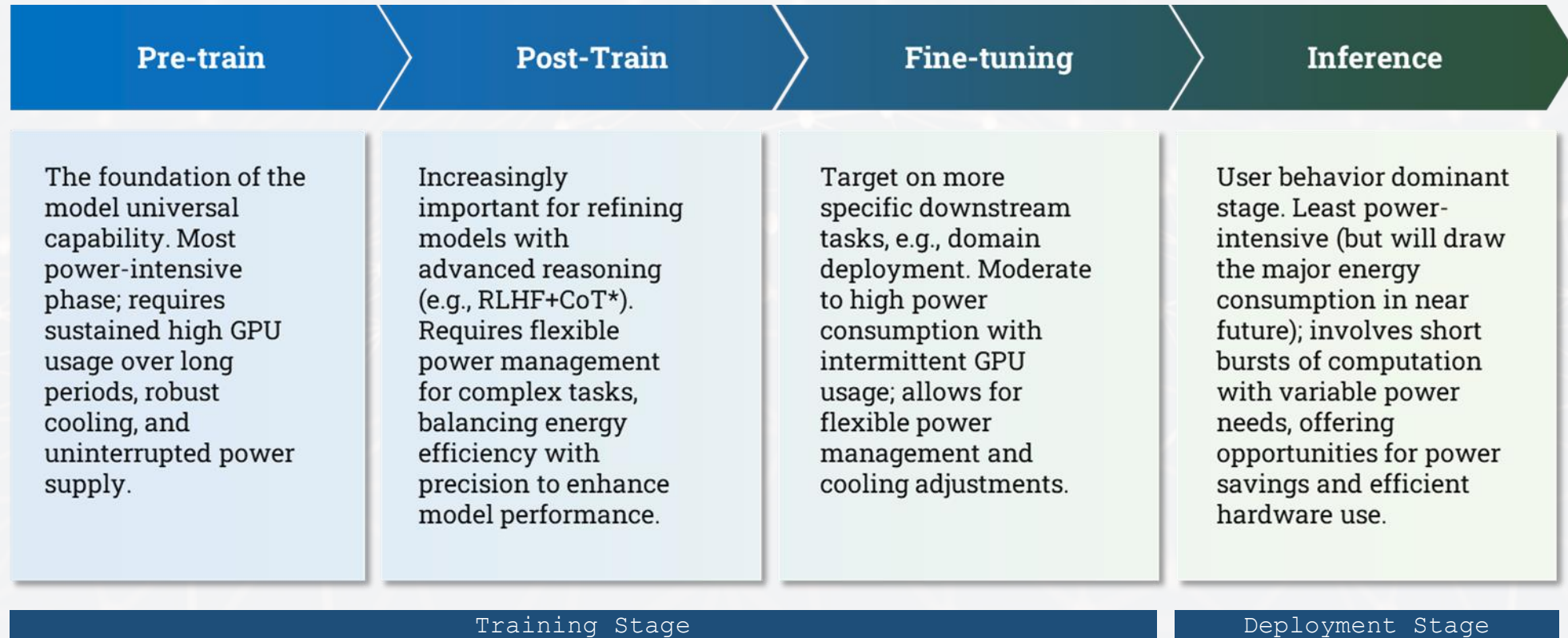
Table 1 – Summary of AI training cluster reference designs

Purpose-built data center optimized for a liquid-cooled cluster. 1,808 kW AI cluster with maximum rack density of 73 kW, using liquid-to-liquid CDUs for heat rejection, optimized with high temperature chillers for improved efficiency.

Attribute	Retrofit design 1	Retrofit design 2	Retrofit design 3	Purpose-built design
AI cluster IT capacity	1,050 kW	904 kW	904 kW	1,808 kW
AI server racks	24 x 40 kW	8 x 73 kW	8 x 73 kW	16 x 73 kW
AI networking racks	6 x 15 kW	8 x 40 kW	8 x 40 kW	16 x 40 kW
Existing IT capacity in room	960 kW	960 kW	960 kW	0 kW
Traditional IT racks	80 x 12 kW	80 x 12 kW	80 x 12 kW	0 kW
Total IT capacity	2,010 kW	1,864 kW	1,864 kW	1,808 kW
AI server rack type	Standard on hard floor	OCP with 3 power shelves on hard floor	OCP with 3 power shelves on hard floor	OCP with 3 power shelves on hard floor
AI server GPU cooling method	Air-cooled	Direct-to-chip	Direct-to-chip	Direct-to-chip
AI cluster heat rejection design	Wide hot aisles with ducted containment, chilled water fan walls, 30C chiller return	Wide hot aisles with ducted containment, liquid-to-air CDUs, chilled water fan walls, 30C chiller return	Wide hot aisles with ducted containment, liquid-to-liquid CDUs, chilled water fan walls, 30C chiller return	Wide hot aisles with ducted containment, liquid-to-liquid CDUs, 2 chiller loops, chilled water fan walls, 40C high temp chiller return
AI server racks power design (per row)	630A busway to redundant tap-offs with redundant 63A 400V rPDUs	800A busway to redundant tap-offs with 3+3 63A 400V power feeds	800A busway to redundant tap-offs with 3+3 63A 400V power feeds	Split 800A busway to redundant tap-offs with 3+3 63A 400V power feeds



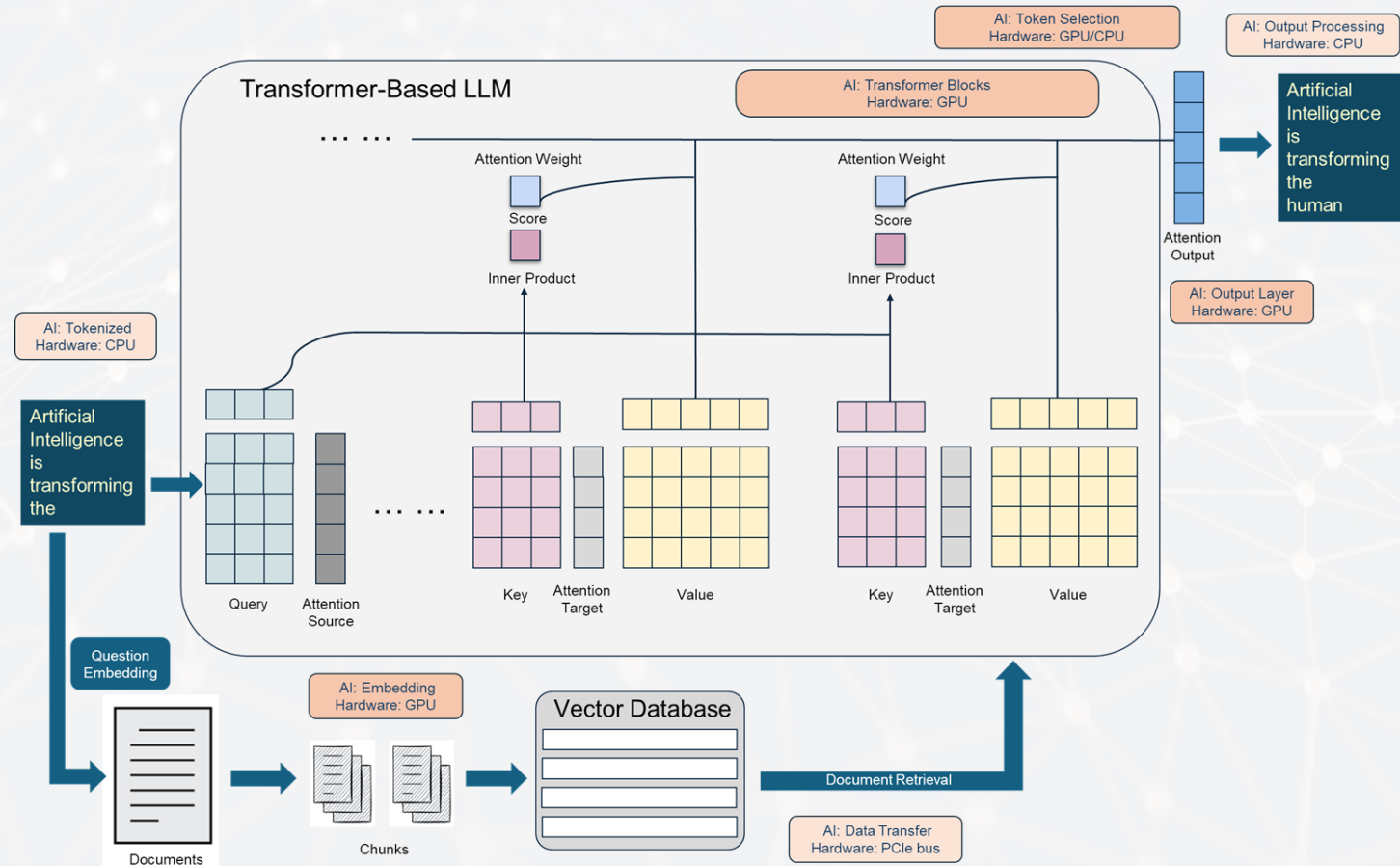
GenAI Pipeline



Transformer for Inference and RAG

The pipeline of a Transformer-based Large Language Model (LLM)

- Transformer training/inference majorly take place in GPUs;
- Retrieval-augmented generation involves document and datasets, communications, and storage.



II - Characteristics of AI Loads

LLM Training Process

Lessons learned from BERT (2020) and Llama 3 (2024)

2020

Number of nodes	Number of V100 GPUs	Time
1 DGX-2	16	33 hr 13 min
4 DGX-2	64	8 hr 41 min
16 DGX-2	256	144 min
64 DGX-2	1,024	44 min

BERT-Large training time using 1 to 64 DGX-2s with Deepspeed*.

Root-cause categorization of unexpected interruptions during a **54-day period** of Llama 3 405B** pre-training. About 78% of unexpected interruptions were attributed to confirmed or suspected hardware issues. (**Total of 466 job interruptions, 47 planned, 419 unplanned, 3 manual intervention**)*

2024

Component	Category	Interruption Count	% of Interruptions
Faulty GPU	GPU	148	30.1%
GPU HBM3 Memory	GPU	72	17.2%
Software Bug	Dependency	54	12.9%
Network Switch/Cable	Network	35	8.4%
Host Maintenance	Unplanned Maintenance	32	7.6%
GPU SRAM Memory	GPU	19	4.5%
GPU System Processor	GPU	17	4.1%
NIC	Host	7	1.7%
NCCL Watchdog Timeouts	Unknown	7	1.7%
Silent Data Corruption	GPU	6	1.4%
GPU Thermal Interface + Sensor	GPU	6	1.4%
SSD	Host	3	0.7%
Power Supply	Host	3	0.7%
Server Chassis	Host	2	0.5%
IO Expansion Board	Host	2	0.5%
Dependency	Dependency	2	0.5%
CPU	Host	2	0.5%
System Memory	Host	2	0.5%

II - Characteristics of AI Loads

LLM Power Consumption Characteristics

Stage	Description	Estimated Power Consumption
Initial Pre-Training	- Cosine learning rate schedule	Very High Due to large-scale training with increasing batch sizes and extensive data processing.
	- Batch size increments	
	- Data mix adjustments	
Long-Context Pre-Training	- Gradual increase of context length to 128K tokens	Extremely High The compute in self-attention layers grows quadratically with sequence length, significantly increasing power consumption.
	- Training on long sequences	
Annealing	- Learning rate reduction to zero	Moderate to High Shorter duration but still computationally intensive due to high-quality data processing and averaging.
	- High-quality data upsampling	
	- Model checkpoint averaging	
Supervised Fine-Tuning (SFT)	- Fine-tuning with supervised data	Moderate Requires less compute than pre-training but still significant due to the size of the model and data.
	- Synthetic data incorporation	
Direct Preference Optimization (DPO)	- Alignment with human preferences	Moderate Computationally intensive due to preference data processing and optimization algorithms.
	- Algorithmic enhancements	
Model Averaging	- Averaging models from various experiments	Low Computational overhead is minimal compared to training stages.
Iterative Rounds	- Multiple cycles of SFT and DPO	Cumulative High Total power consumption accumulates with each iteration, depending on the number of rounds.
	- New data collection	

Outline

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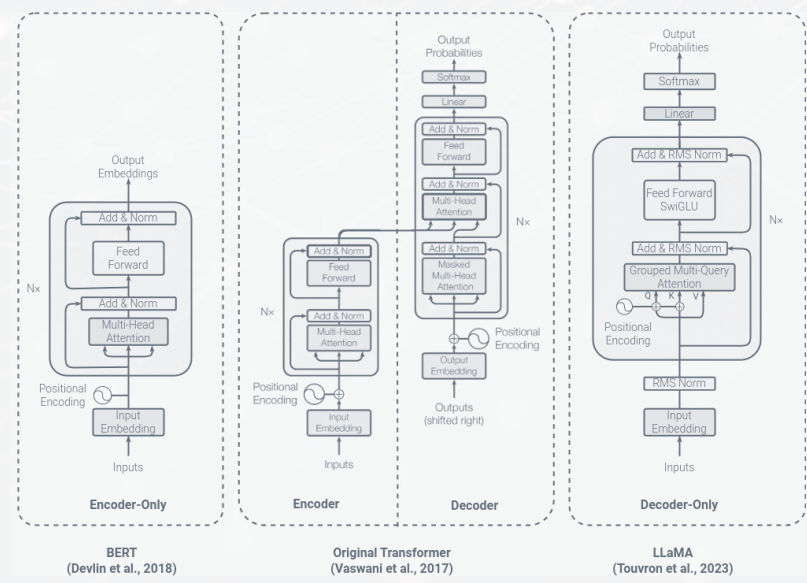
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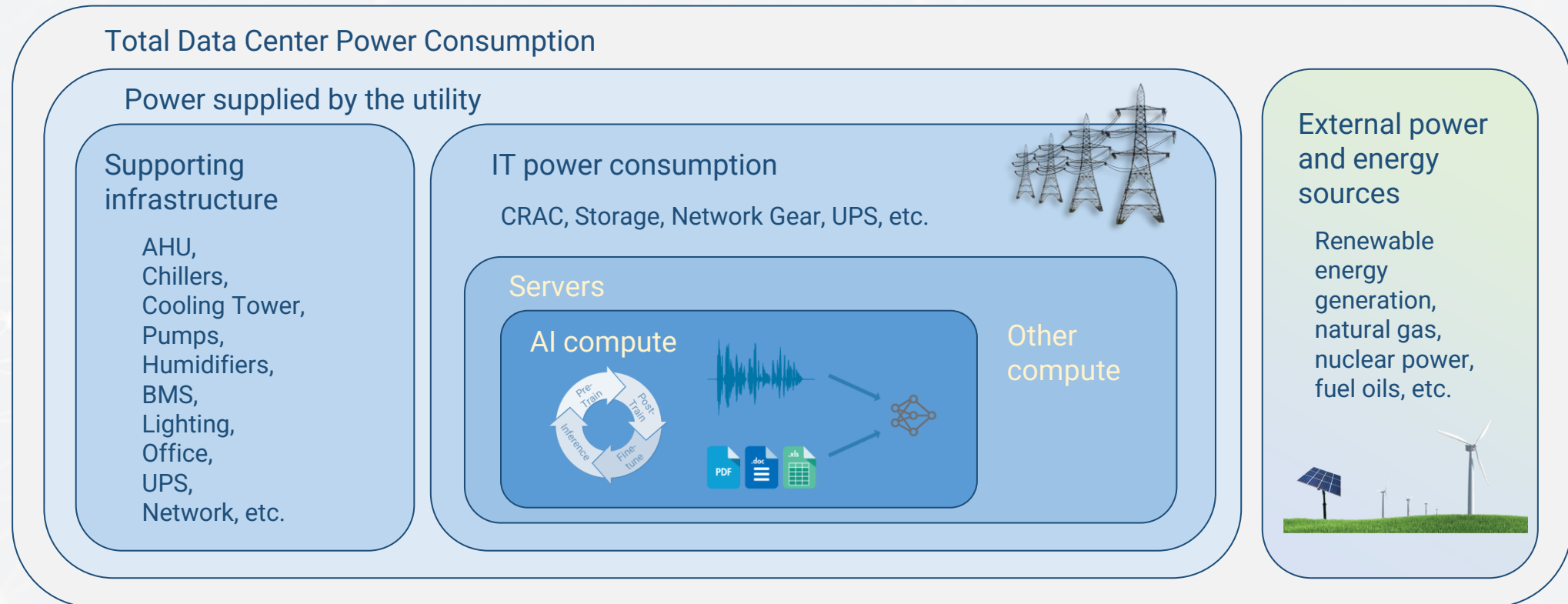
AI Power Consumption Analysis: Case Studies

Future Research Directions

Conclusions



AI-centric Data Center



Layered Data Center Power Consumption

Modeling

Power Consumption Modeling

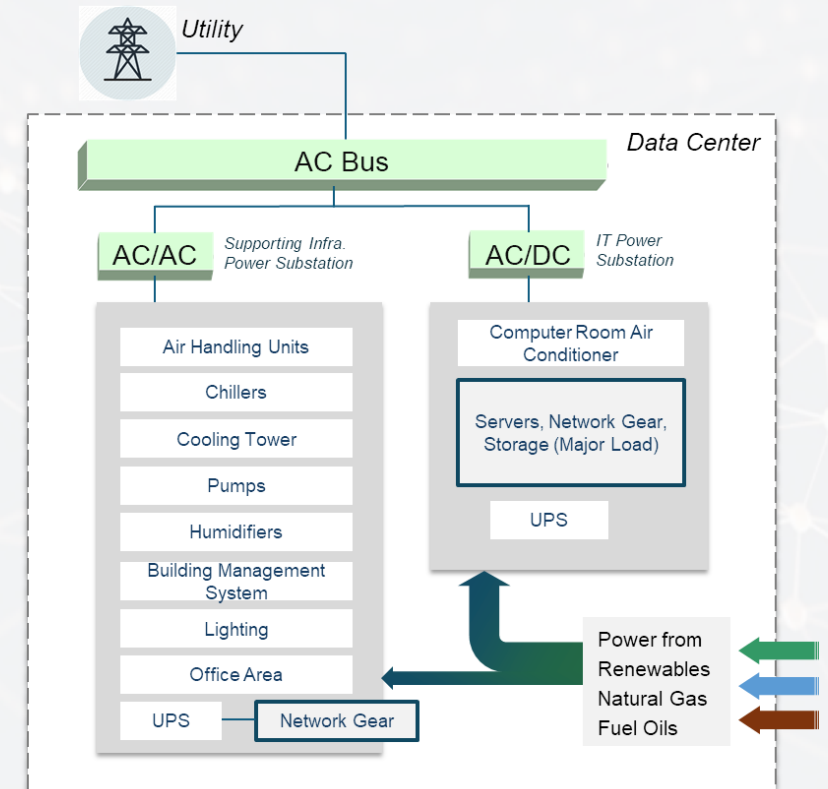
$$P_{\text{total}} = P_{\text{AC_Bus}}$$

$$P_{\text{total}} = P_{\text{Supporting_Infra}} + P_{\text{IT_Power}}$$

$$P_{\text{IT_Power}} = \eta_{\text{AC/DC}} \cdot (P_{\text{Servers}} + P_{\text{NetworkGear}} + P_{\text{Storage}} + P_{\text{CRAC}} + P_{\text{UPS_IT}});$$

$$P_{\text{Supporting_Infra}} = \eta_{\text{AC/AC}} \cdot (P_{\text{AHU}} + P_{\text{Chillers}} + P_{\text{CoolingTower}} + P_{\text{Pumps}} + P_{\text{Humidifiers}} + P_{\text{BMS}} + P_{\text{Lighting}} + P_{\text{Office}} + P_{\text{UPS_Infra}} + P_{\text{Network_Infra}})$$

Decomposition of power models



Modeling

Power Consumption Modeling (steady state)

$$P_{IT_Power} = \eta_{AC/DC} \cdot (P_{Servers} + P_{NetworkGear} + P_{Storage} + P_{ChAC} + P_{UPS_IT});$$

$$P_{Servers} = P_{AI} + P_{Other_Compute}$$

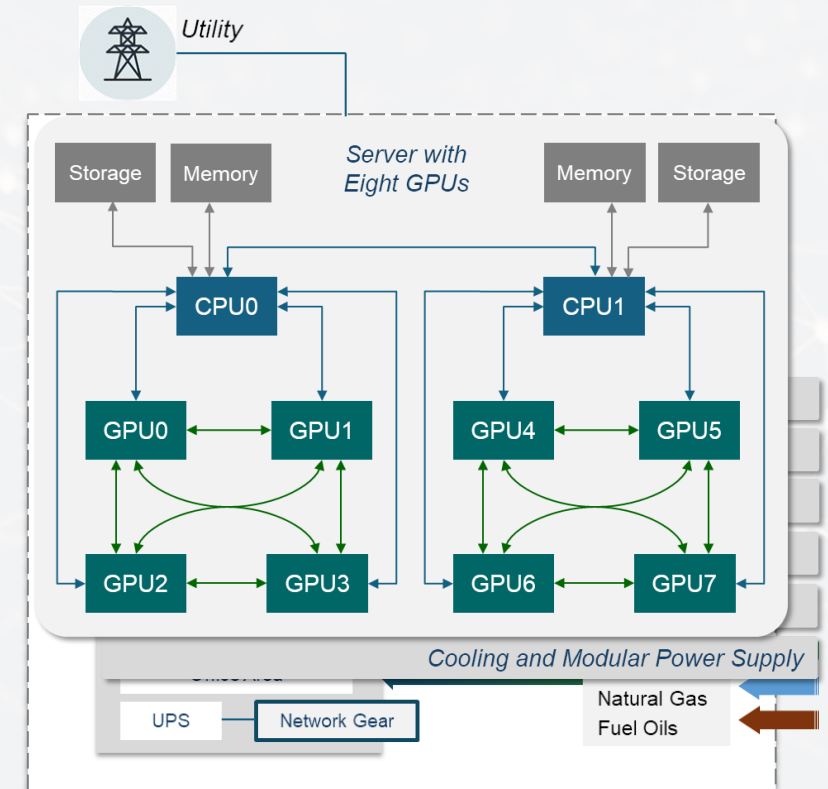
$$P_{AI} \approx N \cdot P_{peak} \cdot U_{phase}$$

Number of AI accelerators

Peak power consumption of a single AI accelerator (this value is normally very close to TDP)

Utilization factor (phase can be pre-train, post-train, fine-tune, or inference)

Decomposition of power models



Modeling

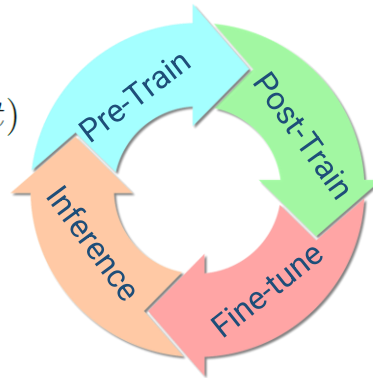
Power Consumption Modeling (dynamic)

$$P_{AI}(t) \approx P_{AI_accelerator}(t) = P_{compute}(t) + P_{memory}(t) + P_{cooling}(t) + P_{auxiliary}(t);$$

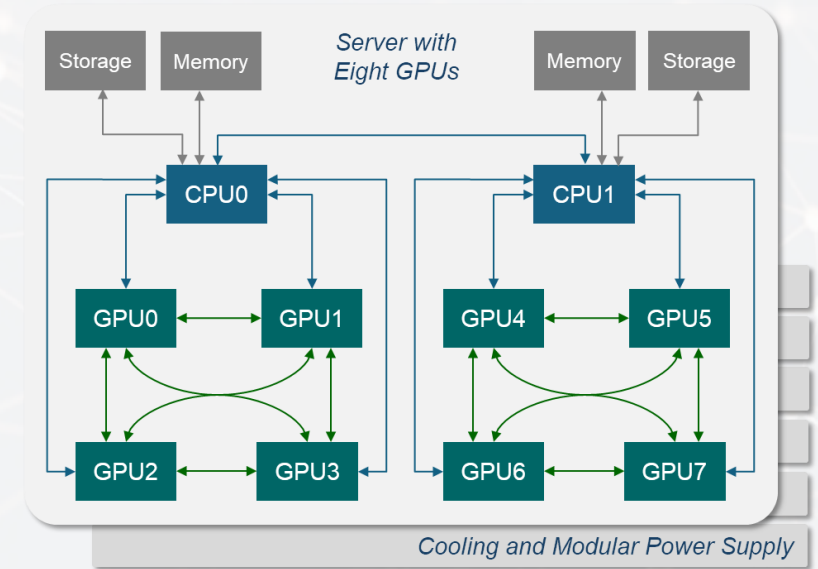
$$P_{training}(t) = P_{base} + \sum_{i=1}^N P_{AI_accelerator_i}(t)$$

$$P_{fine-tuning}(t) = P_{base} + \alpha(t) \cdot \sum_{i=1}^N P_{AI_accelerator_i}(t)$$

$$P_{inference}(t) = P_{base} + \sum_{j=1}^M B_j(t) \cdot P_{query}(s_j, c_j)$$



Different stages

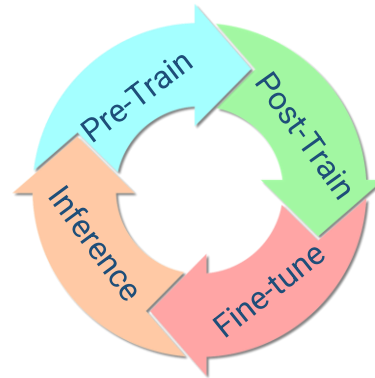


Modeling

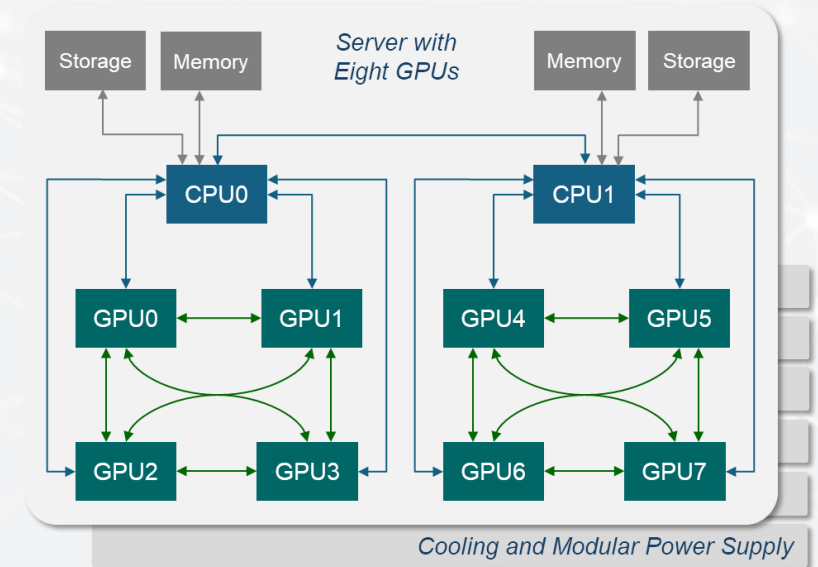
Power Consumption Modeling (dynamic)

$$P_{AI}(t) \approx P_{AI_accelerator}(t) = P_{compute}(t) + P_{memory}(t) + P_{cooling}(t) + P_{auxiliary}(t);$$

$$P_{AI}(t) = w_{train}(t) \cdot P_{training}(t) + w_{finetune}(t) \cdot P_{fine-tuning}(t) + w_{inference}(t) \cdot P_{inference}(t);$$



Different stages



Outline

Background

Characteristics of AI Loads

Power Consumption Modeling of AI Workloads

AI Power Consumption Analysis: Case Studies

Future Research Directions

Conclusions

AI-load behavior

What we look at?

AI running in data center.

AI running locally, e.g., workstation.

For data center

We look at MIT Supercloud with specs: 224 GPU-accelerated nodes, each equipped with two 20-core Intel Xeon Gold 6248 processors (384GB RAM) and two NVIDIA Volta V100 GPUs (32GB RAM each), as well as additional CPU-only nodes.

For local workstation

We look at two setups as detailed in the table.

Setup	Hardware	Task Type	Tasks
Setup1	Nvidia 4090 1000W PSU	Pre-train	GPT2-124M training
		Fine-tuning	-
		Inference	-
Setup2	AMD 7900XTX 750W PSU	Pre-train	nanoGPT training
		Fine-tuning	GPT2-medium fine-tuning
		Inference	Llama 3.2 1B inference
			Llama 3.2 3B inference
			Mamba inference
			Transformer inference

LLM Case Study Setups, Hardware, and Tasks

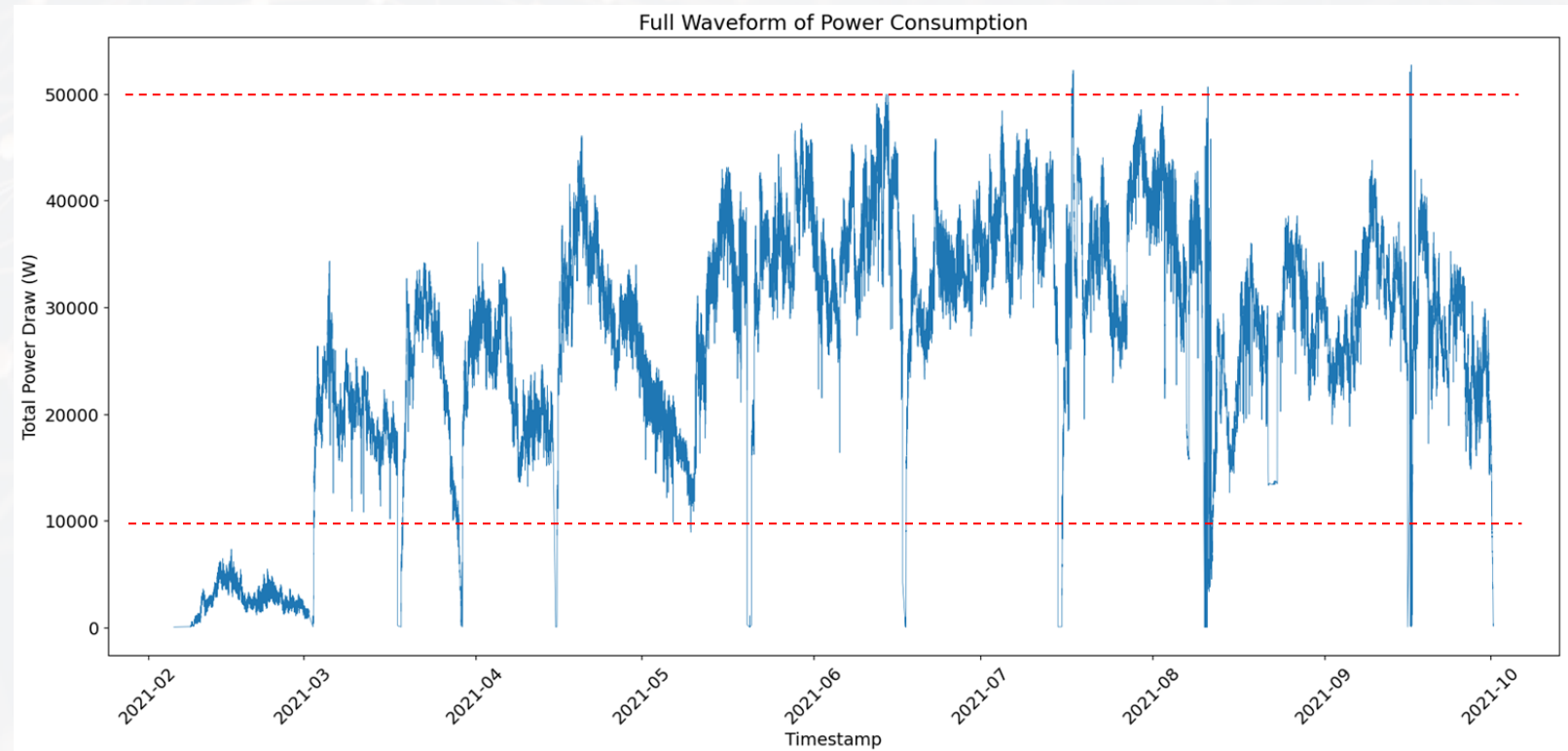
Data center behavior

The spec.

224 GPU-accelerated nodes, each equipped with two 20-core Intel Xeon Gold 6248 processors (384GB RAM) and two NVIDIA Volta V100 GPUs (32GB RAM each), as well as additional CPU-only nodes.

The 7-month GPU power consumption of data center:

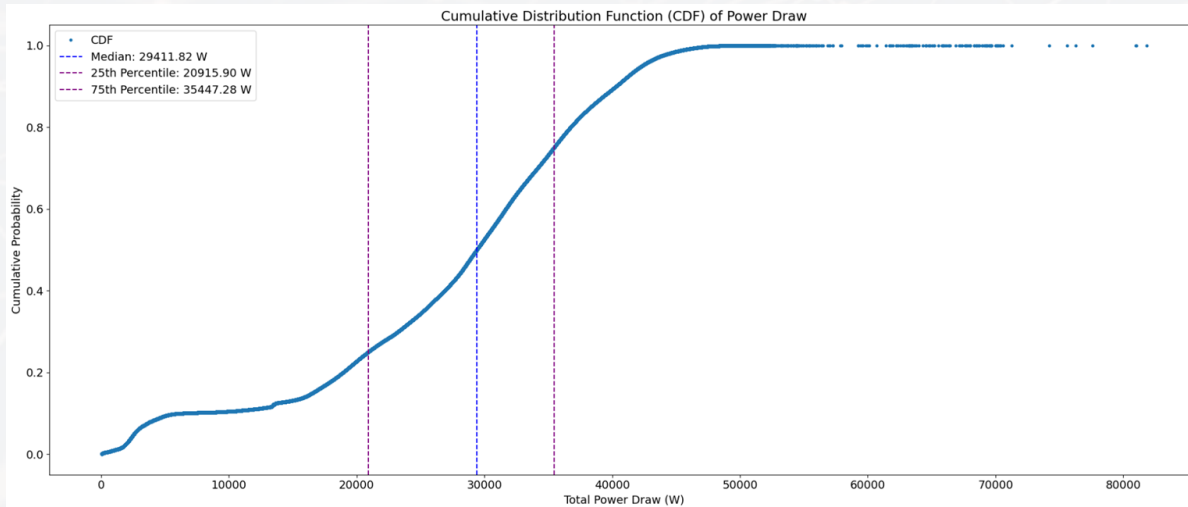
- Dynamic range: near 0W to >50kW
- Short power spikes
- Baseline power: When active, system rarely drops below ~10kW



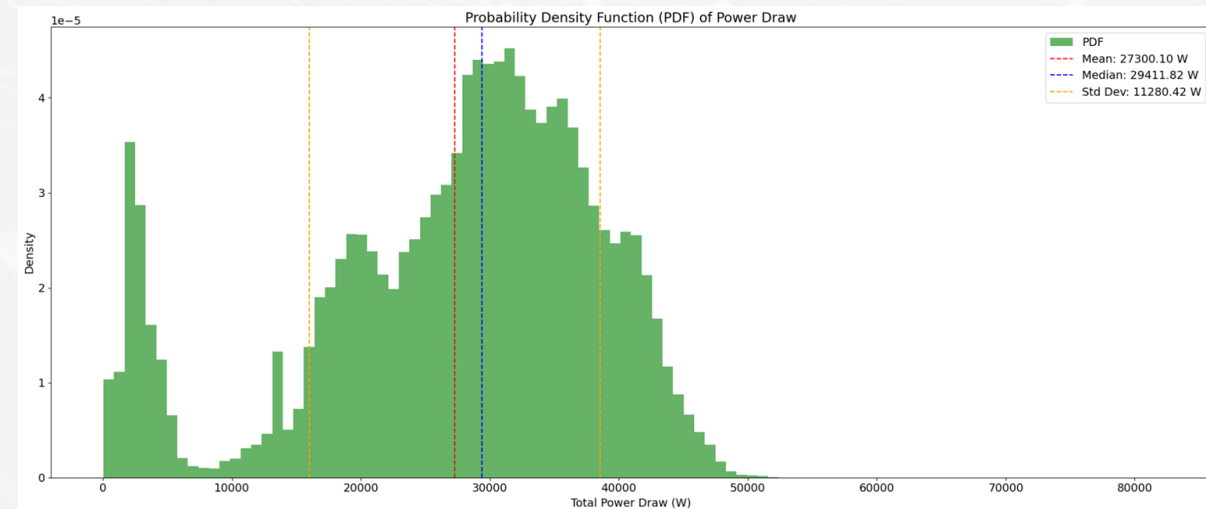
Data center behavior

Statistical insights: CDF and PDF

- **Median power:** 29,411.82W, indicating typical operation uses about 50-60 GPUs at full capacity
- **Wide distribution**
- **Smooth curve:** Indicates a continuous range of power states rather than just a few discrete levels



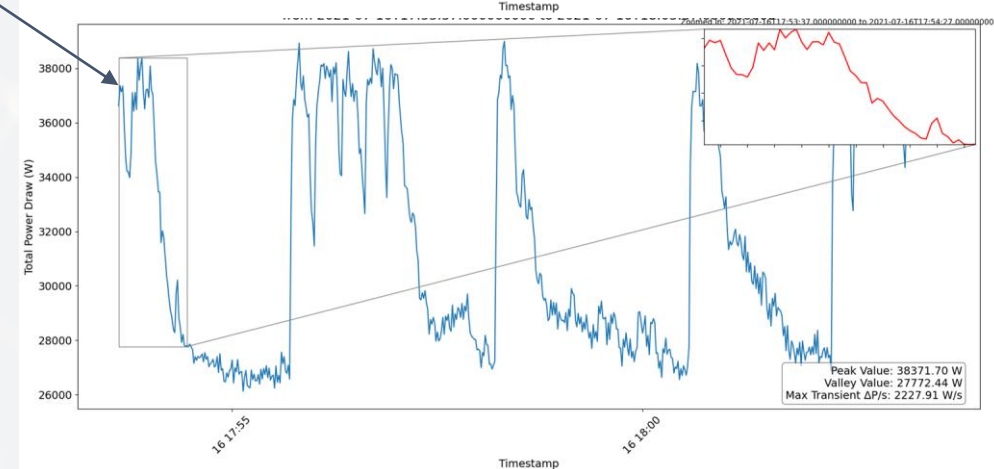
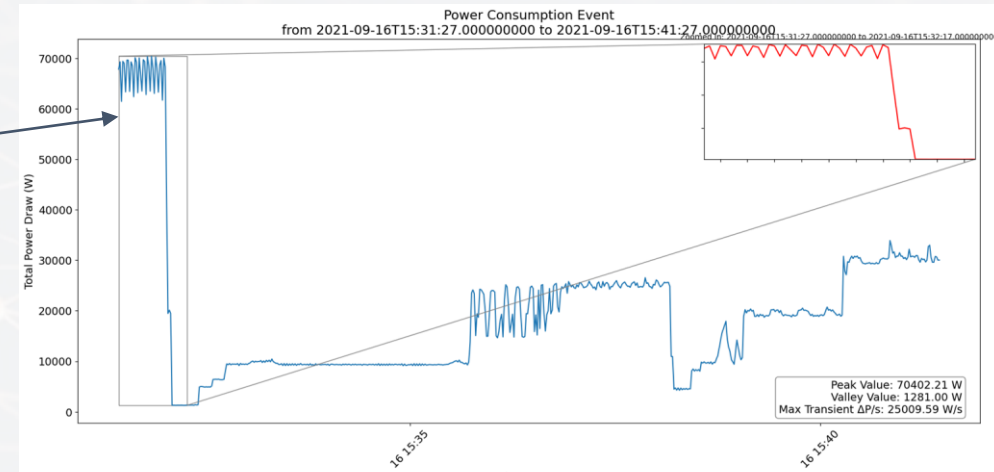
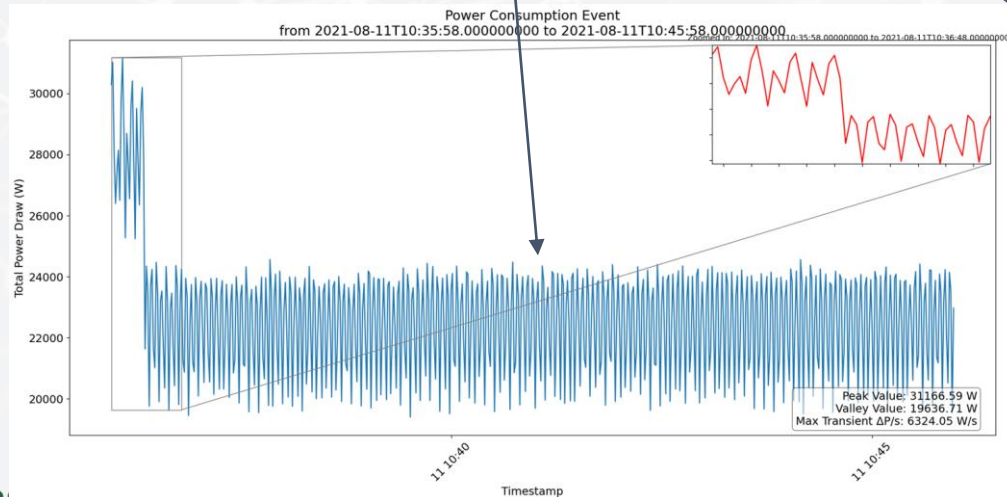
- **Bimodal distribution:** Two distinct peaks in power usage
- **Low power mode:** Peak around 5kW, idle or minimal operation state
- **High power mode:** Peak around 30kW, typical high utilization
- **Long upper tail:** Extends to ~80kW, indicating occasional periods of extreme demand



Data center behavior

Noticeable moments (3 out of 612 events)

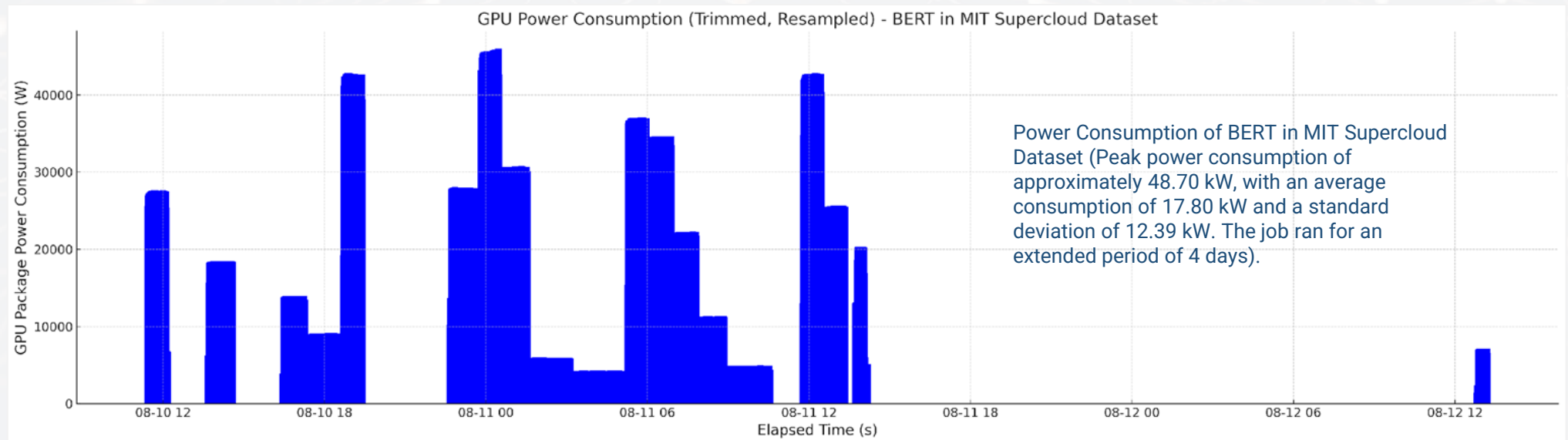
- Dramatic power swings from 0W to 70kW
- Sharp spikes of high power usage, followed by quick returns to baseline
- Regular cycles between ~20kW and ~24kW



Data center behavior

The jobs*:

Vision Networks	Job Count	Language Models	Job Count	Graph Neural Networks	Job Count
VGG	560	Bert	189	DimeNet	33
ResNet	464	DistillBert	172	SchNet	39
Inception	484			PNA	27
U-Net	1431			NNConv	32

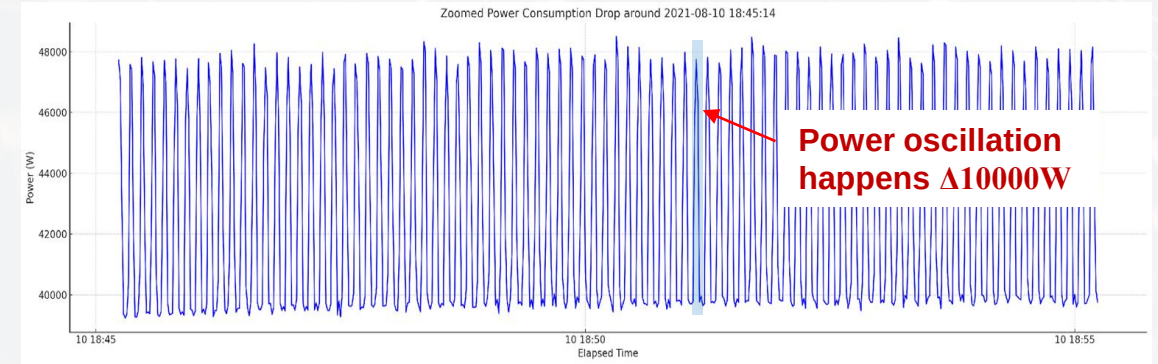
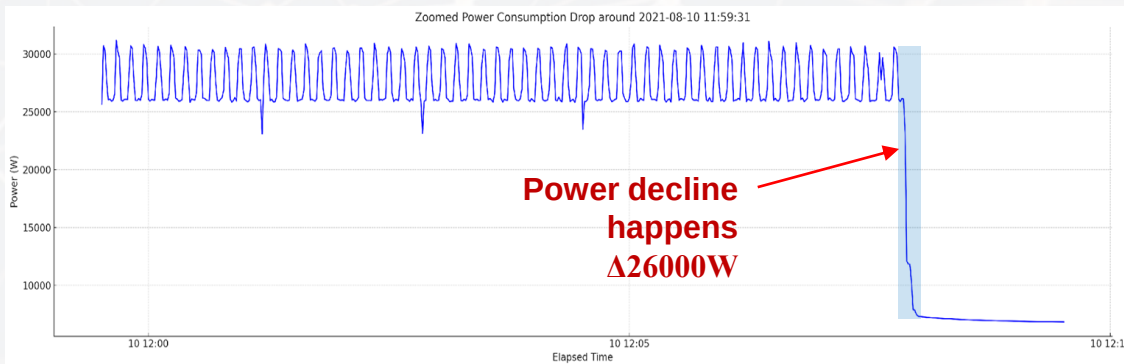
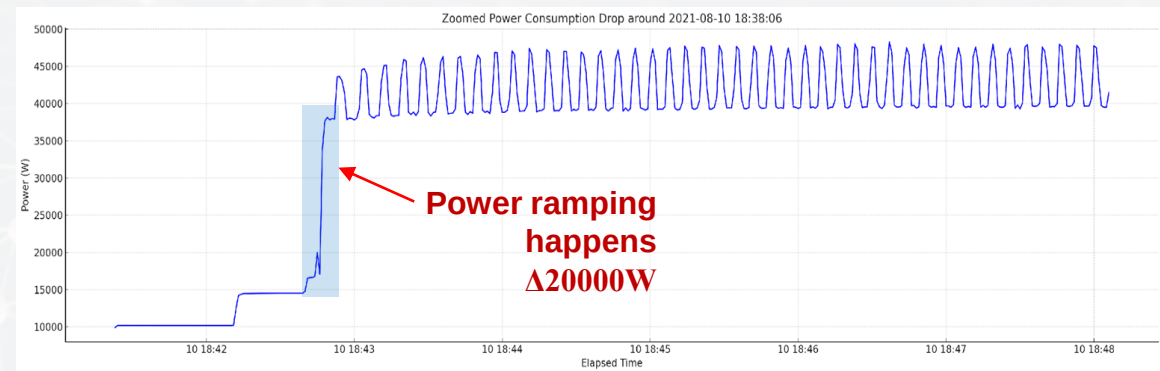


IV - AI Power Consumption Analysis: Case Studies

Data center behavior

The observation

Mean Power Consumption: 19,567.37 W
Standard Deviation: 13,693.69 W
Minimum Power Consumption: 467.26 W
Maximum Power Consumption: 51,593.23 W



Baselines: LLMs

Training performance:

- GPT2-124M trained using Setup 1 (1000W).
- nanoGPT trained using Setup 2 (750W).

Fine-tuning performance:

- GPT2-medium fine-tuned using Setup 2 (750W).

Inference performance:

- Llama 3.2 1B/3B, Mamba, and Transformer models tested for inference using Setup 2 (750W).

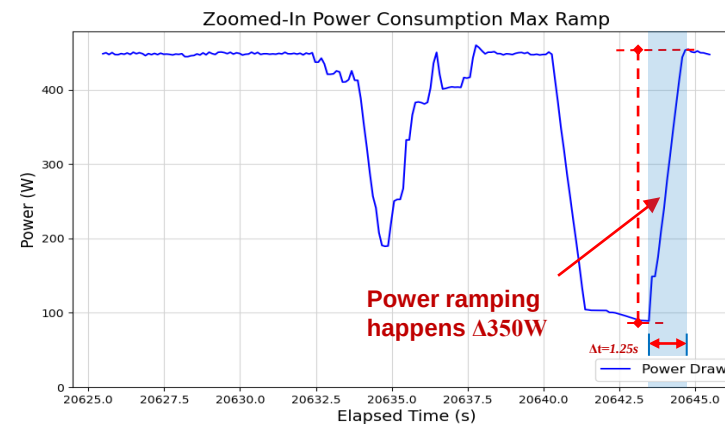
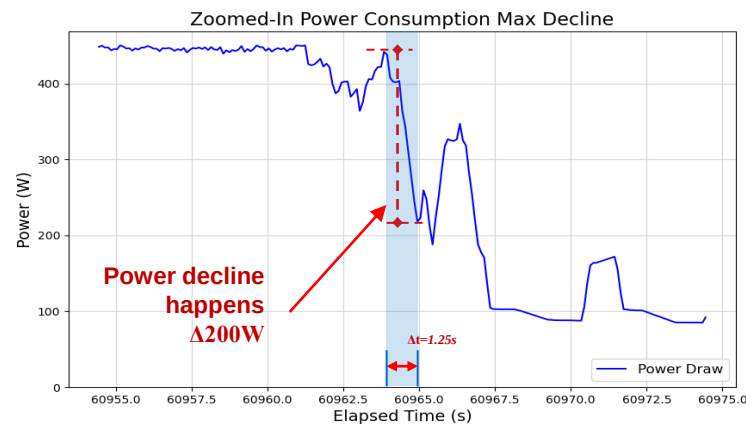
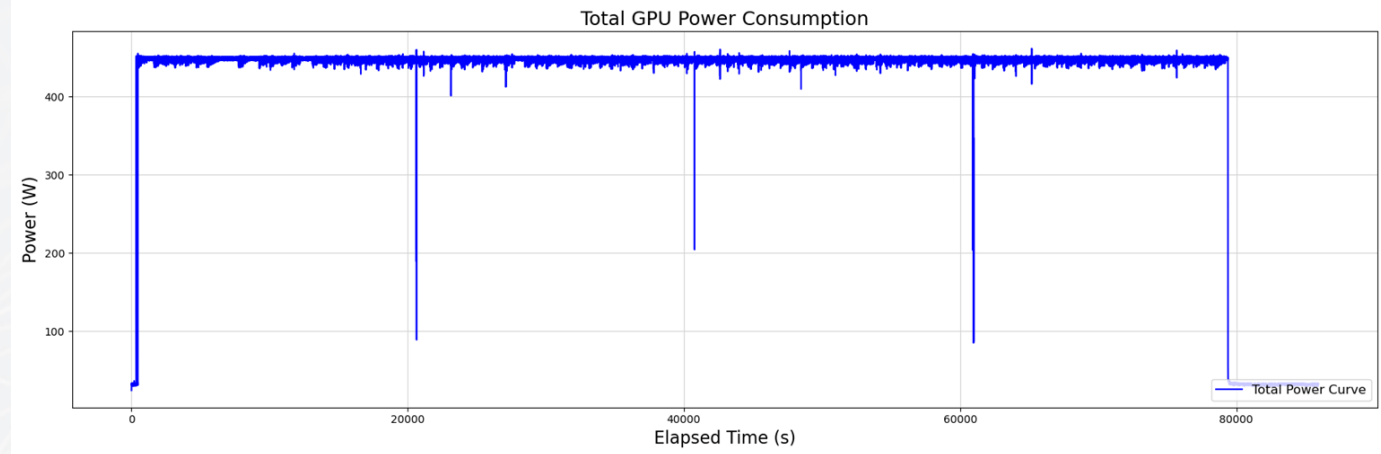
Series	Model Name	Parameters	Layers	Hidden Size	Attention Heads	Context Length
GPT-2	GPT-2 Small	124M	12	768	12	1024
	GPT-2 Medium	355M	24	1024	16	1024
	GPT-2 Large	774M	36	1280	20	1024
	GPT-2 XL	1.5B	48	1600	25	1024
nanoGPT	nanoGPT (base)	Varies	Varies	Varies	Varies	Configurable
	nanoGPT-124M	124M	12	768	12	Configurable
Mamba	mamba-130M	130M	24	768	N/A	2048
	mamba-370M	370M	48	1024	N/A	2048
	mamba-790M	790M	48	1536	N/A	2048
	mamba-1.4B	1.4B	48	2048	N/A	2048
	mamba-2.8B	2.8B	64	2560	N/A	2048
GPT-Neo	GPT-Neo 125M	125M	12	768	12	2048
	GPT-Neo 1.3B	1.3B	24	2048	16	2048
	GPT-Neo 2.7B	2.7B	32	2560	20	2048
	GPT-NeoX 20B	20B	44	6144	64	2048
Llama 3.2	Llama 3.2 1B	1.23B	N/A	N/A	N/A	128k
	Llama 3.2 3B	3.21B	N/A	N/A	N/A	128k

Specifications of Various AI Model Series: Small Language Model

LLM training

GPT-2 124M training (NVIDIA RTX 4090)

- High, stable power consumption averaging 414W, peaking at 461W
- 22-hour training period with consistent power draw
- Full waveform shows a plateau-like pattern with occasional sharp dips

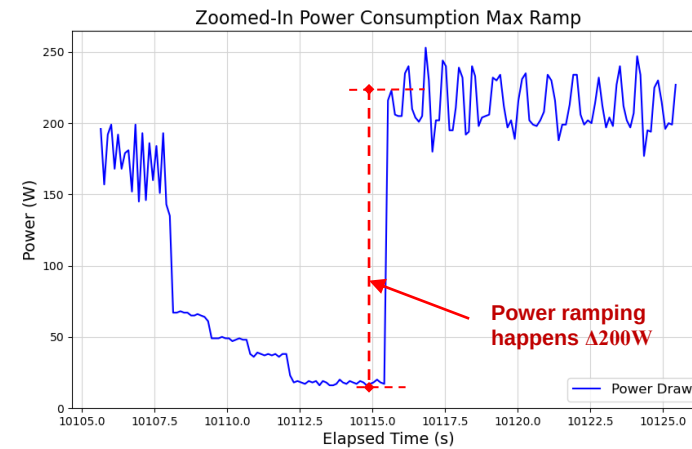
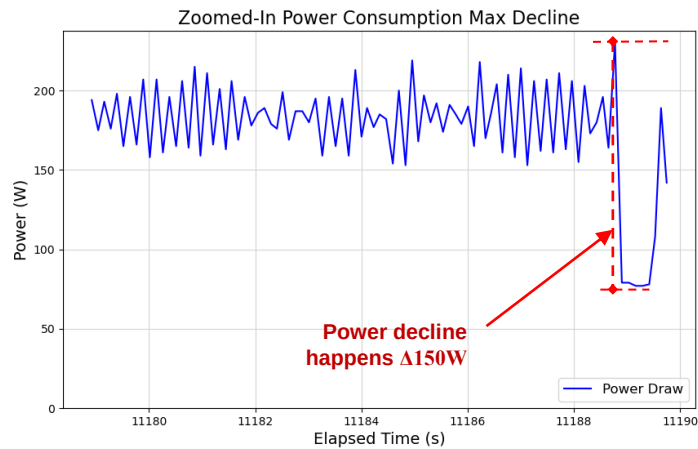
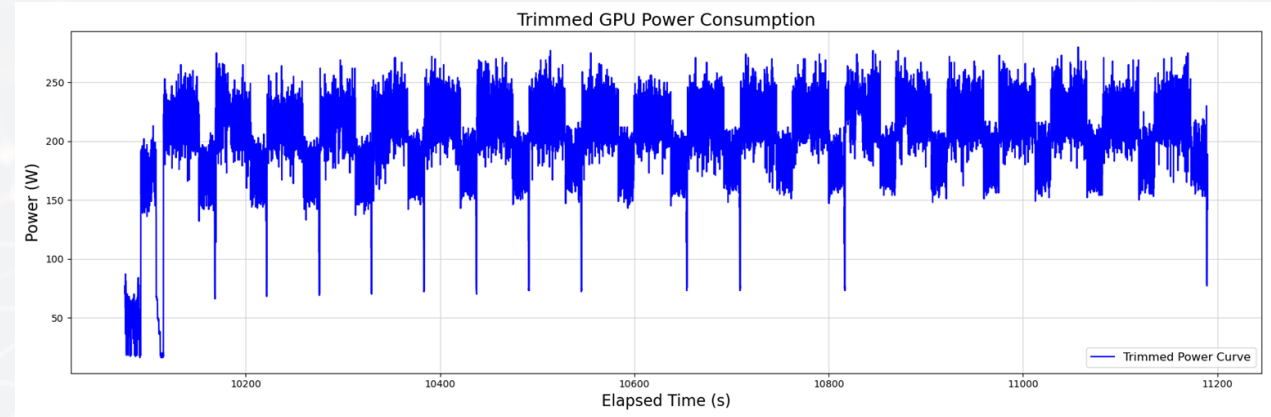


IV - AI Power Consumption Analysis: Case Studies

LLM training

nanoGPT training (AMD Radeon RX 7900 XTX)

- Variable power consumption ranging from approximately 50W to 250W
- Frequent fluctuations in power draw throughout the training session
- Full waveform exhibits a rhythmic pattern with regular power cycles



IV - AI Power Consumption Analysis: Case Studies

Fine-tuning of LLM

Initialization and Setup Phase (0-350s):

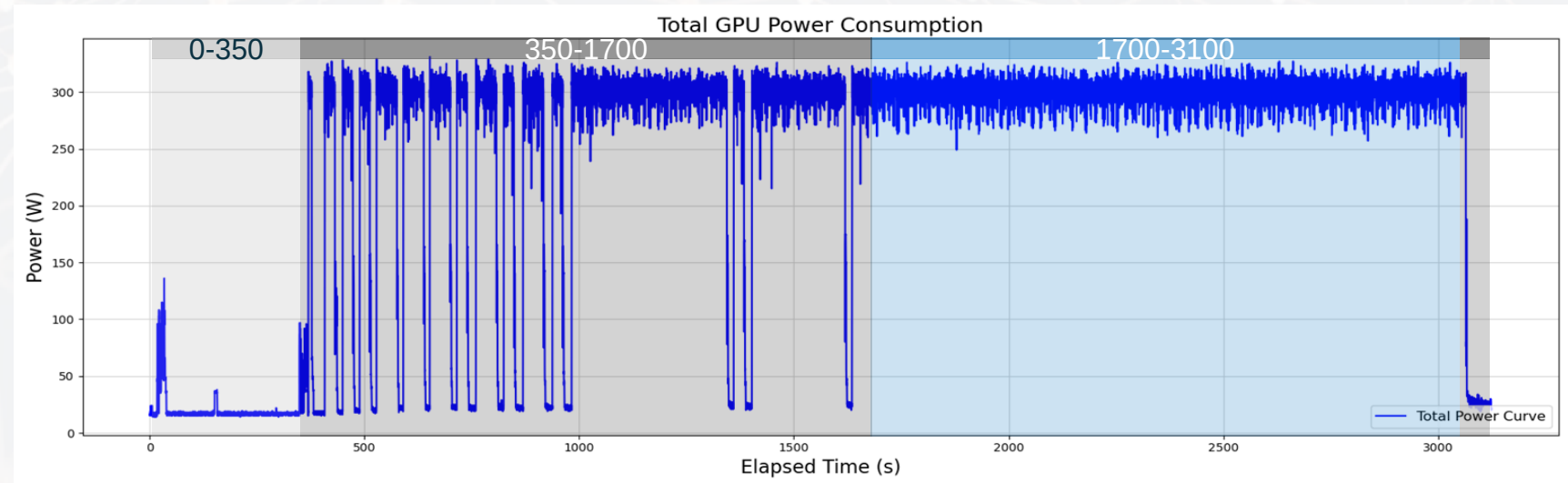
- Initial spike in power consumption
- Followed by a period of low power draw
- Activities: Model initialization, data loading, optimizer setup
- Power pattern reflects memory allocations, weight loading, and initial data processing

Early Training and Learning Rate Warm Up Phase (350-1700s):

- High and fluctuating power consumption
- Power peaks reaching around 330W
- Regular sharp drops, likely corresponding to evaluation intervals

Late Training and Learning Rate Decay Phase (1700-3100s):

- Power consumption remains high but appears slightly more consistent
- Learning rate enters decay phase
- Differences in fluctuation patterns compared to early training



IV - AI Power Consumption Analysis: Case Studies

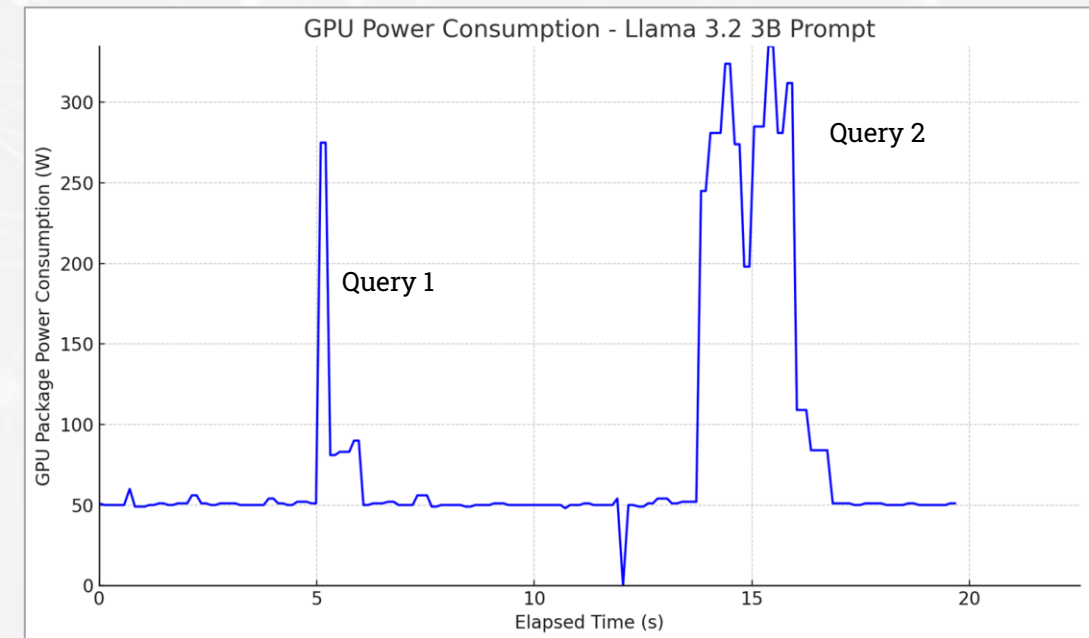
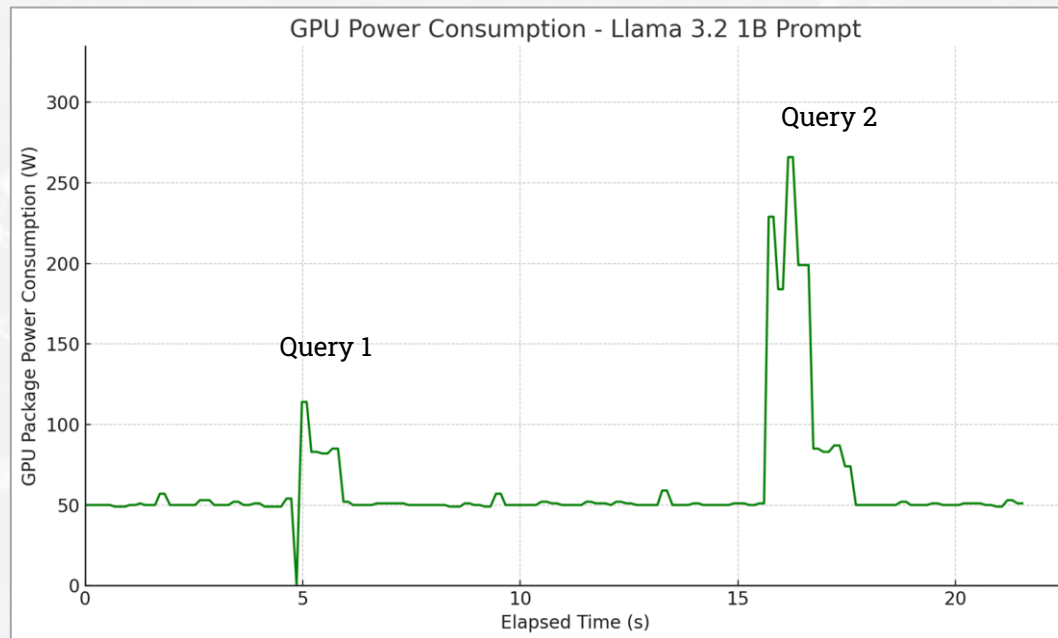
Llama 3.2 1B and 3B

Training Time (370k/460k GPU
hours, 240 tons CO2eq)

LLM Inference with different models

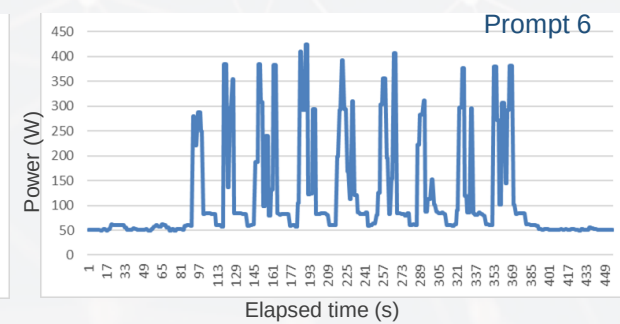
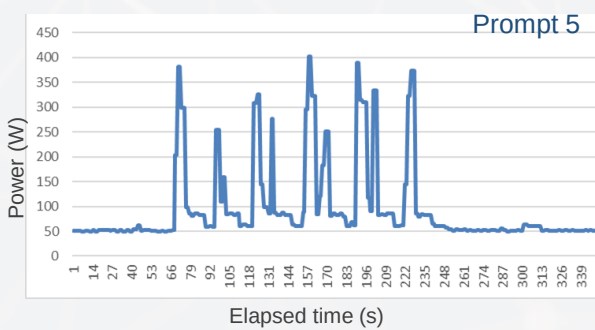
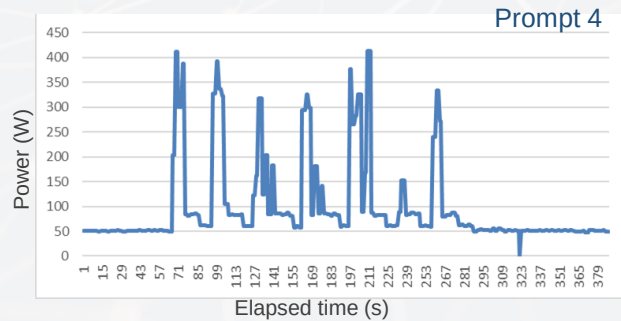
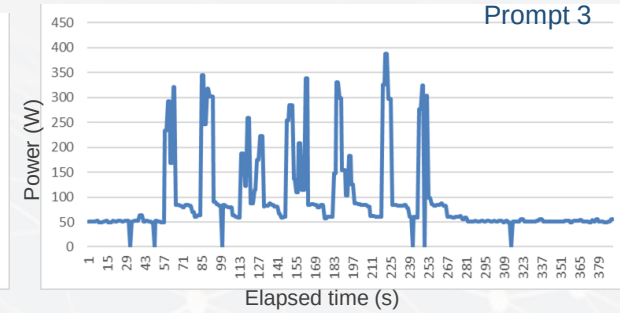
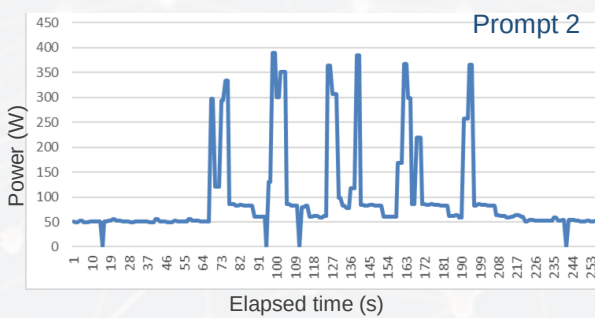
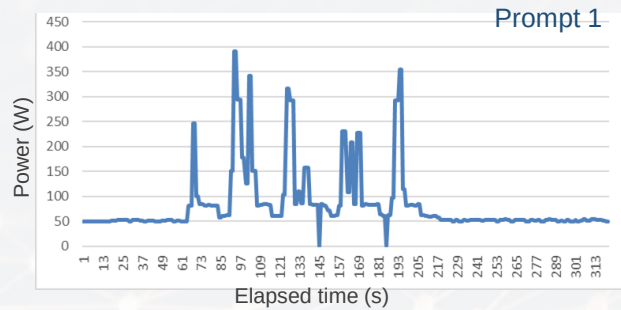
Query 1: What countries start with the letter V?

Query 2: I'm Suzie and I'm 6 years old. My mommy says Santa will visit us in December and give us prezzies and candy but my older brother Johnny says that there is no Santa Claus. Is Santa Claus real?



LLM Inference with Chain-of-Thought

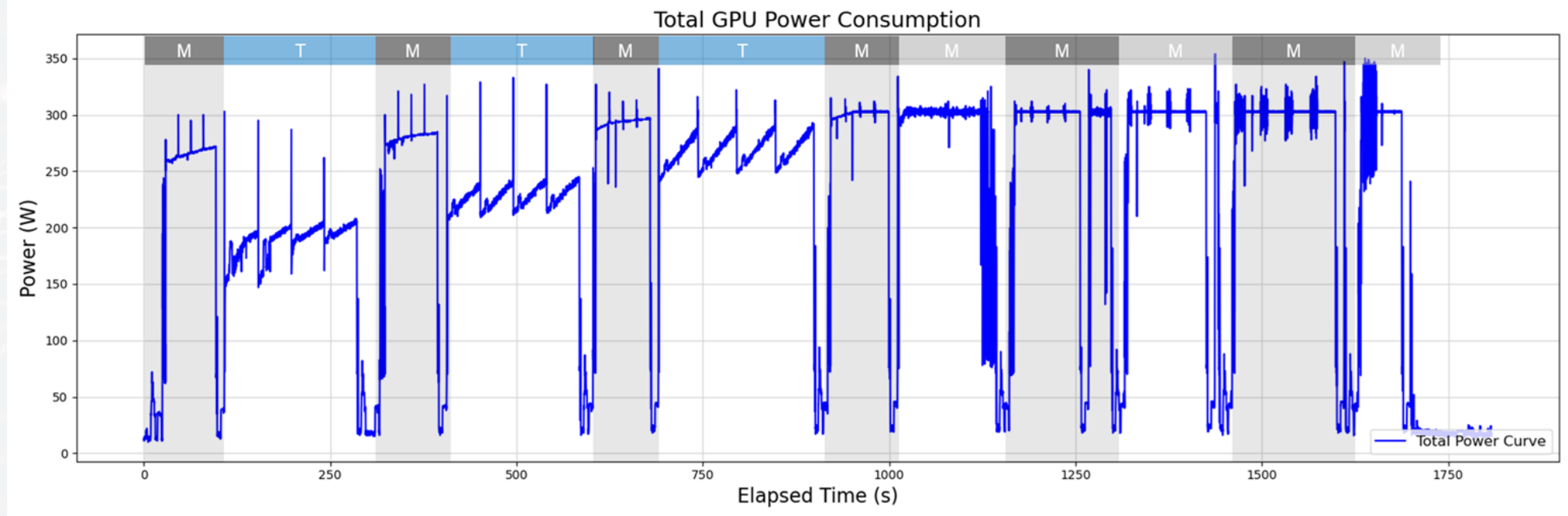
Llama 3.1 8B model (with *Chain of Thought* process running locally on AMD Radeon RX 7900 XTX)



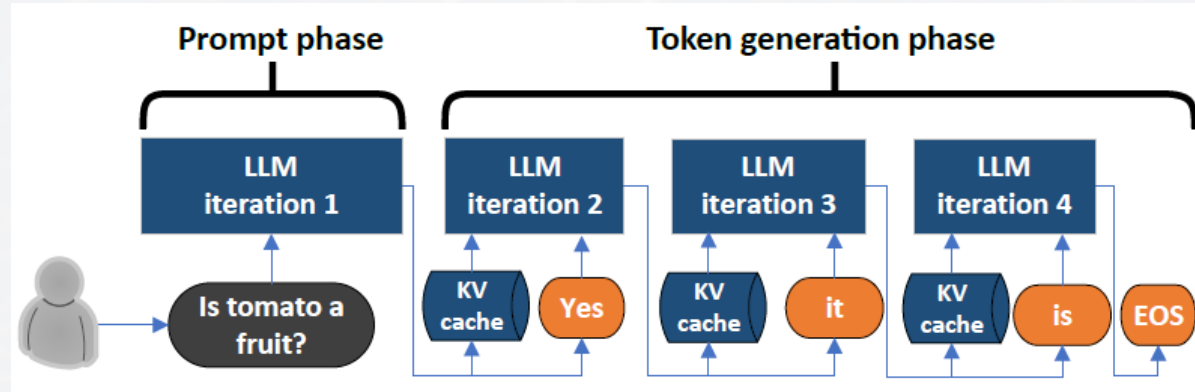
LLM inference with different batch sizes

Mamba (state-spaces/mamba-2.8b) and Transformer (EleutherAI/gpt-neo-2.7B) models

- Both models show similar power consumption patterns at lower batch sizes
- As batch sizes increase, power consumption patterns diverge
- Mamba model continues operating at larger batch sizes where the Transformer model fails



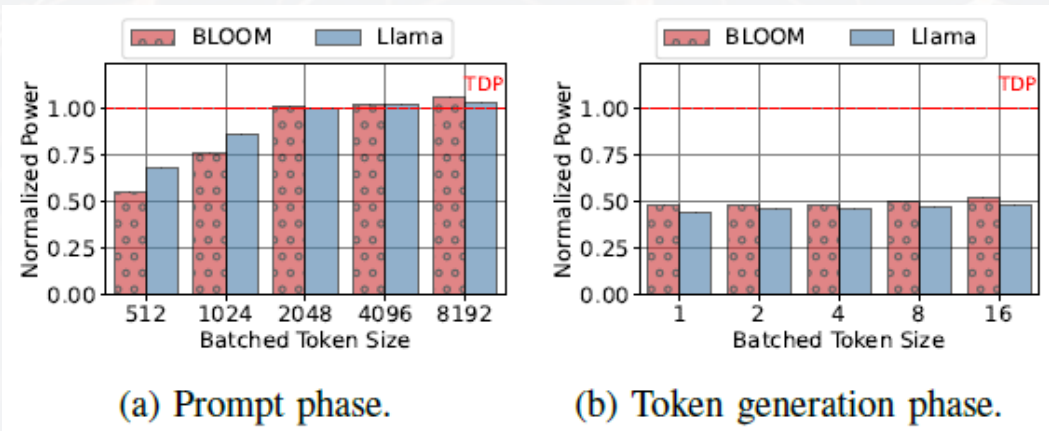
More details in LLM inference



An example of the prompt and token phases in a GPT (decoder) model.

GPU power draw normalized to the thermal design power (TDP) when running prompt and token generation phases.

Model	#Layers	Hidden size	#Heads
Llama2-70B	80	8192	32
BLOOM-176B	70	14336	112



- **Prompt phase** is compute intensive, its power draw increases with batch size.
- **The token phase** is memory bound and its power draw does not vary when increasing the number of tokens to process.

Insights

Dynamics:

- LLM training workloads exhibit rapid and significant **power fluctuations** due to alternating computation- and communication-intensive phases across GPUs. LLM inference tasks also present rapid power fluctuations, with power surges reaching peaks comparable to training but over shorter durations.

Software side:

- Model **architecture** and **size** significantly impact power consumption patterns in LLM training. Larger and more complex models result in higher and more stable power draws, operating near the GPU's maximum capacity.
- Smaller or efficiency-focused models show more variable power consumption with frequent fluctuations, affecting power management strategies. Distinct training phases, such as initialization, learning rate adjustments, and evaluation intervals, exhibit unique power consumption patterns.
- Phases like learning rate warmup cause high and fluctuating power consumption, while evaluation periods lead to **sharp drops** to near-idle power levels.

Hardware side:

- Each GPU can experience power swings of several hundred watts **within seconds** during LLM training iterations.
- Differences between GPU manufacturers lead to varied power consumption behaviors during LLM training.
- Variations in GPU architecture and power management affect average power draw and the magnitude of power transients.
- To handle these rapid power fluctuations, data center clusters need specialized power infrastructure capable of managing swift and substantial power changes without compromising stability.

Outline

Background

Characteristics of AI Loads

Power Consumption Modeling of AI Workloads

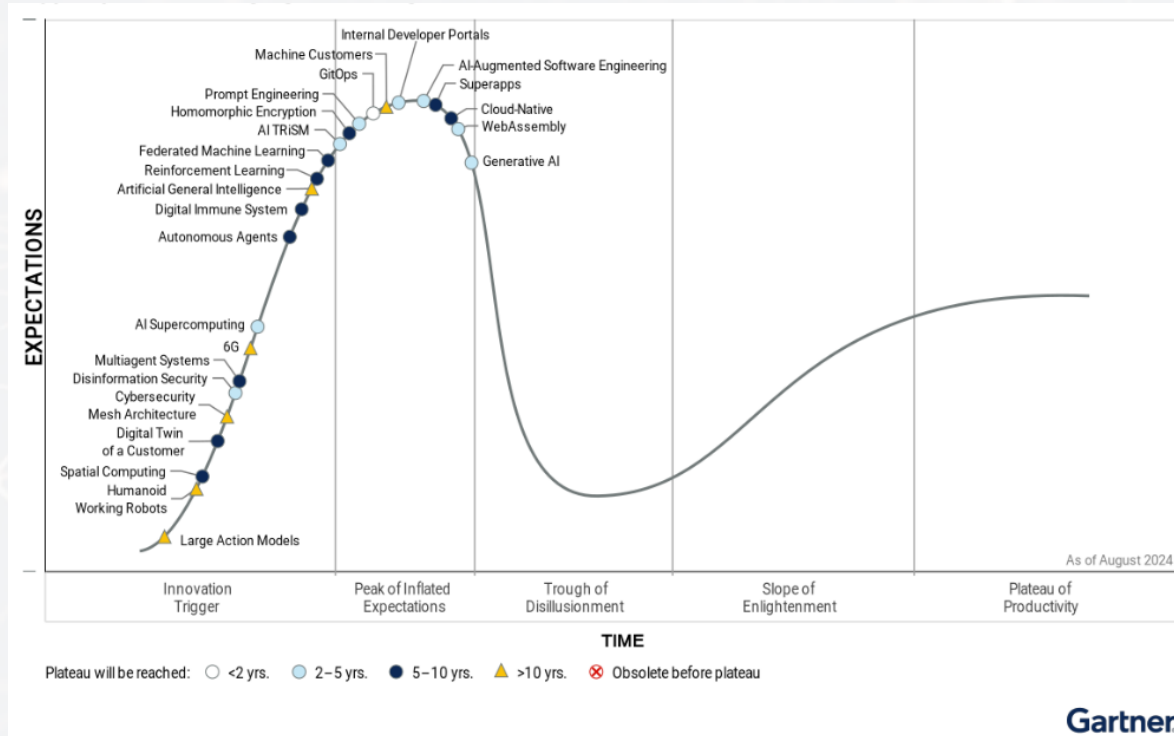
AI Power Consumption Analysis: Case Studies

Future Research Directions

Conclusions

Need more AI?

Hype Cycle for Emerging Technologies, 2024



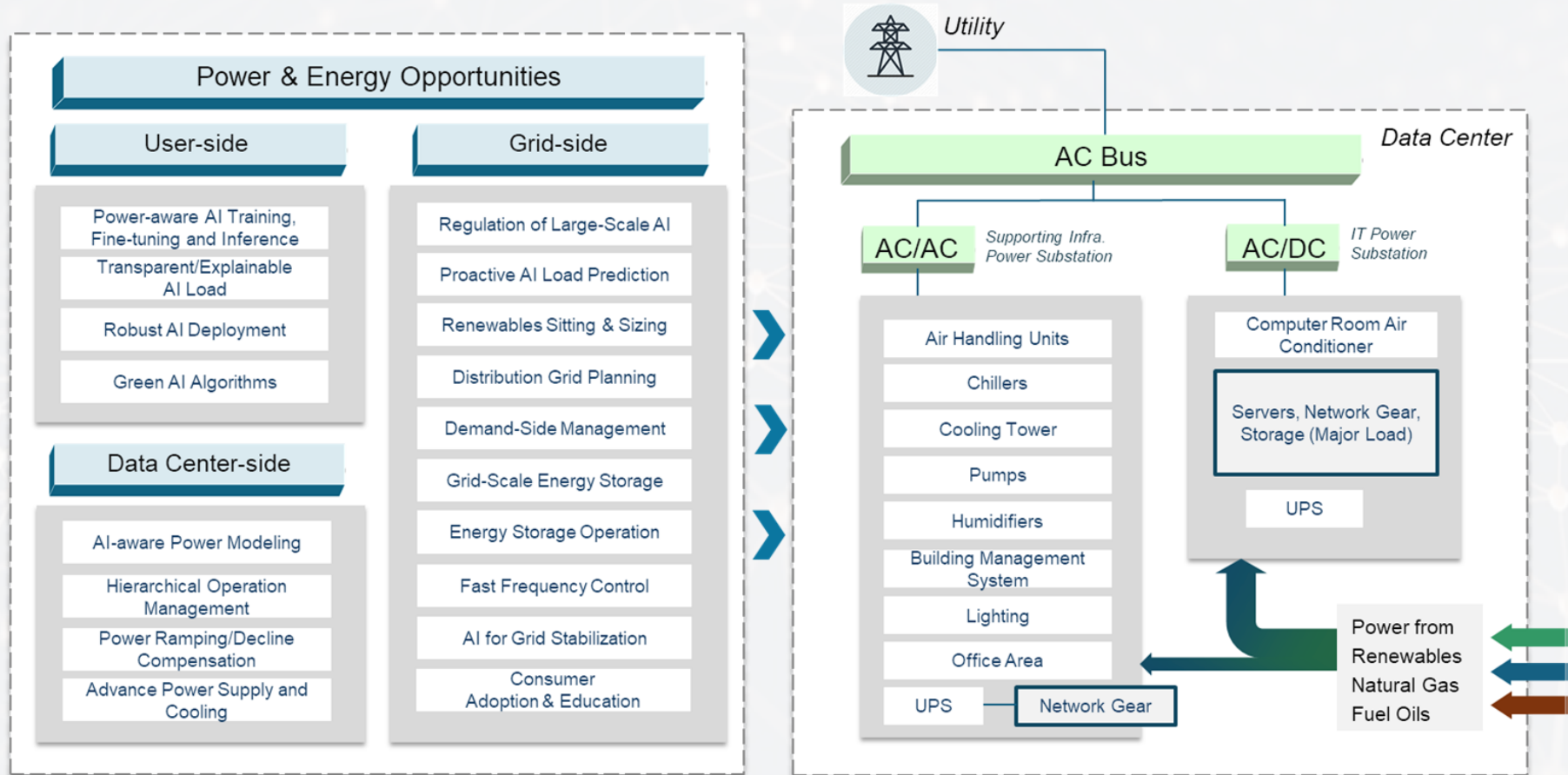
Snapshot of AI regulation and regulatory proposals in 2024

REGION	FOCUS	EXAMPLES OF POLICIES/LAWS
U.S. federal	AI risk assessment	■ Algorithmic Accountability Act (H.R. 5628; S.2892) ■ DEEP FAKES Accountability Act (H.R. 3220) ■ Digital Services Oversight and Safety Act (H.R. 6796)
	AI Bill of Rights	■ White House's Blueprint for an AI Bill of Rights
	AI framework	■ NIST's AI Risk Management Framework
U.S. state and city	AI regulation	■ California, Connecticut, Texas, Illinois, Colorado, NYC are among states and cities with laws or bills to regulate AI
Global	High-risk AI applications	■ European Union's Artificial Intelligence Act
	Generative AI regulation	■ China's proposed Administrative Measures for Generative Artificial Intelligence Service
	Risk mitigation, transparency	■ Canadian Parliament's Artificial Intelligence and Data Act
	Developing regulations	■ At least eight other countries across the Americas and Asia

More AI is on the road

- AI market size is expected to reach **\$407 billion** by 2027.
- AI will have an estimated **21%** net increase on the United States GDP by 2030.
- **Half** of U.S. mobile users use voice search every day.
- AI is expected to see an annual growth rate of **37.3%** from 2023 to 2030.
- A **quarter** of companies are adopting AI because of labor shortages.
- Over **60%** of business owners say AI will increase productivity.
- **65%** of consumers say they'll still trust businesses who use AI.

From AI4Grid to Grid4AI



AI-User Perspective

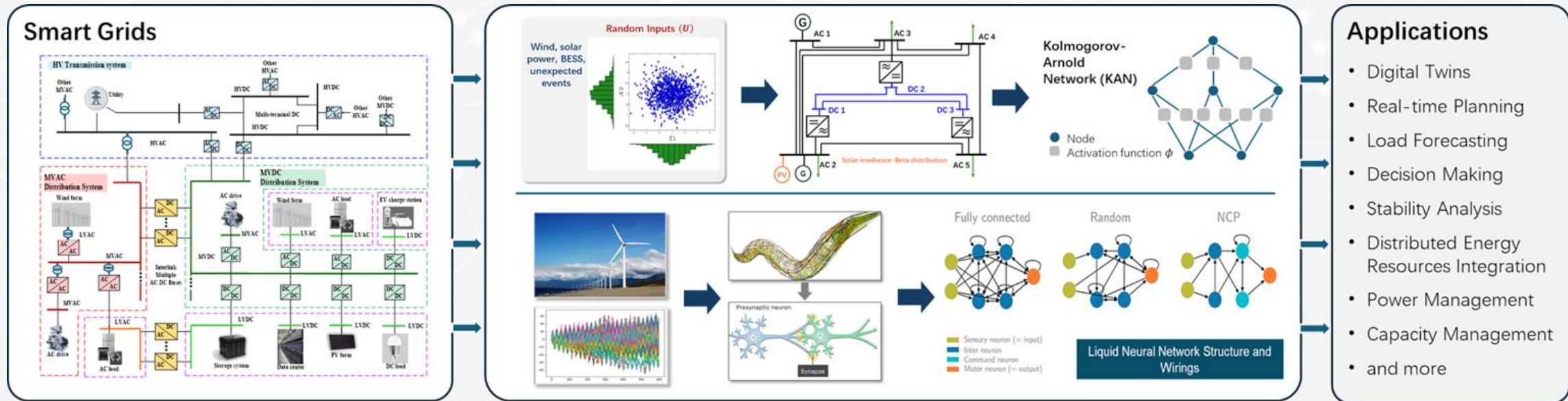
Power-aware AI Training, Fine-tuning, and Inference

Green AI Algorithms

Transparent/Explainable AI Load

Robust AI Deployment

Example: Efficient AI for Smart Grids



Data-center Perspective

AI-aware Power Pattern Modeling

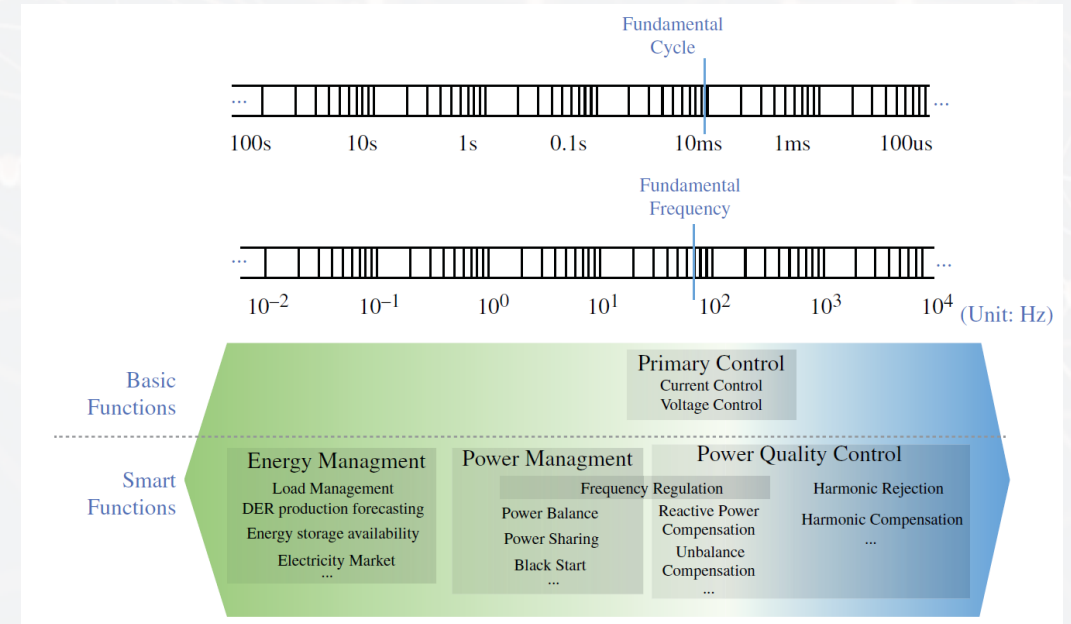
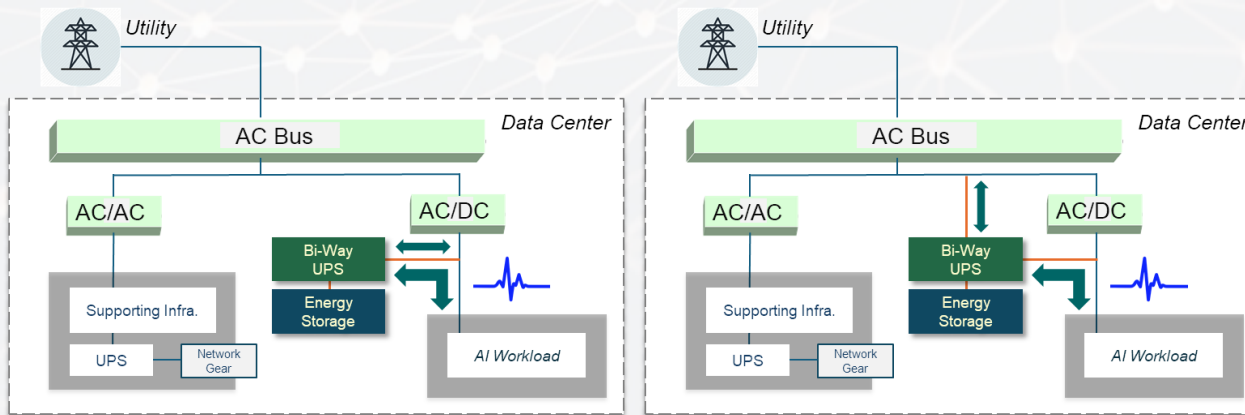
Power Ramping/Decline Compensation

Hierarchical Operation Management

Advanced Power Supply and Cooling

Example: Operation management over wide time scale

Example: Multifunctional Power Electronics Device (smart UPS)



Functions of a smart microgrid.

Grid-side

Regulation of Large-Scale AI

Proactive AI Load Prediction

Renewables Siting and Sizing

Distribution Grid Planning

Demand-Side Management

Grid-Scale Energy Storage

Fast Frequency Control

AI for Grid Stabilization

Consumer Adoption and Education

- Regulation of AI Power Consumption: Developing frameworks to manage AI's unique power consumption across **training, fine-tuning, and inference phases**, including advance notifications and power caps during peak hours.
- AI Load Prediction Models: Creating machine learning models to predict AI-related power consumption patterns, focusing on **rapid fluctuations** during inference and sustained high-power usage during training.
- Renewable Energy Integration: Exploring optimal strategies for **colocating** AI data centers with renewable energy sources, **aligning** AI workloads with peak renewable generation to reduce emissions and energy costs.
- Grid Planning for AI Workloads: Developing new tools to account for the clustered nature of AI operations and their **impact** on local grid infrastructure and stability.
- Demand-Side Management: Designing demand response programs tailored to AI operations, **optimizing** power-intensive tasks during low-demand periods or excess renewable generation.
- Grid-Scale Energy Storage: Innovating **hybrid** storage systems to manage AI's variable power demands, including long-term storage for training phases and fast-response storage for inference fluctuations.
- Fast Frequency Control: Investigating **ultra-fast** response techniques, potentially using AI accelerators for real-time frequency regulation and synthetic inertia.
- AI for Grid Stabilization: Leveraging AI infrastructure during idle periods to **support** grid services such as voltage support, reactive power compensation, and real-time power flow management.
- Consumer Education and Adoption: Developing energy efficiency **ratings** and **educational** programs for AI-powered consumer devices to promote sustainable AI use and inform consumers about energy implications.

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Summary

- **AI as a Unique Power Grid Load:** The analysis identifies AI, particularly large language models (LLMs), as a distinctive load on power grids, differing significantly from traditional data center loads.
- **Extreme Dynamics of AI Loads:** AI infrastructure, especially LLMs, introduces extreme dynamics in power consumption that have not been seen before, including:
 - **Ultra-Low Inertia:** Rapid changes in power consumption that occur with little to no delay.
 - **Subsecond Peak-Idle Fluctuation:** Dramatic shifts between peak power usage and idle states within very short time frames (subseconds).
 - **Unpredictable Interruptions:** Power fluctuations caused by interruptions during AI training sessions.
 - **User-Behavior Dominant Inference Power Consumption:** Inference power consumption is highly dependent on user activity and behavior, making it difficult to predict.
- **Challenges to Power Grids:** These unique characteristics collectively present significant challenges to the stability and management of power grids, which are traditionally designed for more predictable loads.
- **Urgent research** is needed to ensure power grids can sustain AI's growth while maintaining reliability and sustainability.
 - **AI-Specific Power Management:** Developing advanced control systems to handle AI's erratic power demands.
 - **Innovations in Power Electronics:** Improving converters (e.g., SiC, GaN) and hybrid AC/DC systems to enhance efficiency and stability.
 - **Energy Storage Integration:** Optimizing hybrid energy storage (flywheels, supercapacitors) to buffer AI-induced power fluctuations.
 - **AI-Driven Predictive Models:** Using AI to forecast power usage and improve grid reliability.

Live Q&A

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Yuzhuo Li PhD, Postdoc
Email: yuzhuo@ualberta.ca

The webinar is based on an ongoing project (*Yuzhuo Li, Mariam Mughees, Yize Chen, Yunwei Ryan Li*) which is hosted at https://github.com/chennnnnyize/LLM_Impact_Energy_Systems